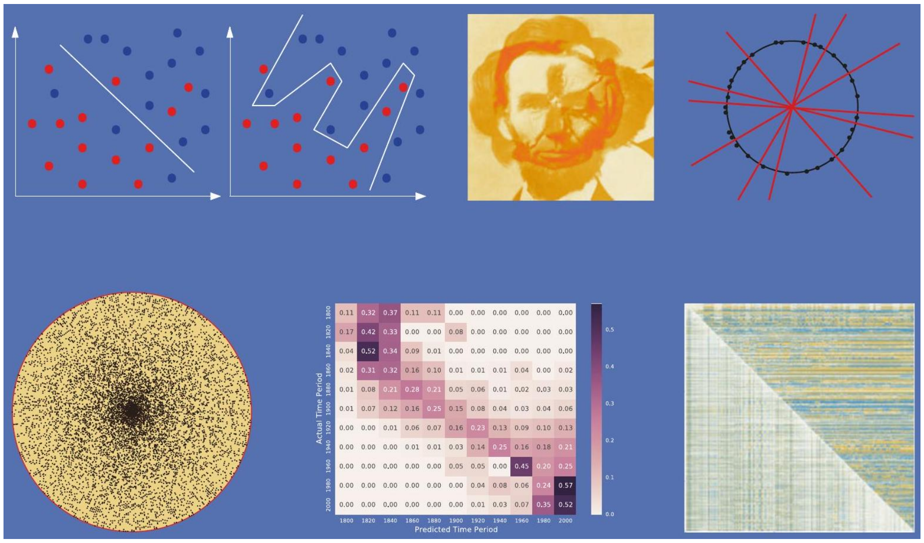
# Contents

# Texts in Computer Science

Statistical Distributions

TEXTS IN COMPUTER SCIENCE THE Data Science Design MANUAL

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# Chapter 1

# Preface

Making sense of the world around us requires obtaining and analyzing data from our environment. Several technology trends have recently collided, providing new opportunities to apply our data analysis savvy to greater challenges than ever before.

Computer storage capacity has increased exponentially; indeed remembering has become so cheap that it is almost impossible to get computer systems to forget. Sensing devices increasingly monitor everything that can be observed: video streams, social media interactions, and the position of anything that moves. Cloud computing enables us to harness the power of massive numbers of machines to manipulate this data. Indeed, hundreds of computers are summoned each time you do a Google search, scrutinizing all of your previous activity just to decide which is the best ad to show you next.

The result of all this has been the birth of *data science*, a new field devoted to maximizing value from vast collections of information. As a discipline, data science sits somewhere at the intersection of statistics, computer science, and machine learning, but it is building a distinct heft and character of its own. This book serves as an introduction to data science, focusing on the skills and principles needed to build systems for collecting, analyzing, and interpreting data.

My professional experience as a researcher and instructor convinces me that one major challenge of data science is that it is considerably more subtle than it looks. Any student who has ever computed their grade point average (GPA) can be said to have done rudimentary statistics, just as drawing a simple scatter plot lets you add experience in data visualization to your resume. But meaningfully analyzing and interpreting data requires both technical expertise and wisdom. That so many people do these basics so badly provides my inspiration for writing this book.

**To the Reader** I have been gratified by the warm reception that my book *The Algorithm Design Manual* [
citeskiena2008algorithm] has received since its initial publication in 1997. It has been recognized as a unique guide to using algorithmic techniques to solve problems that often arise in practice. The book you are holding covers very

different material, but with the same motivation.

In particular, here I stress the following basic principles as fundamental to becoming a good data scientist:

- *Valuing doing the simple things right*: Data science isn't rocket science. Students and practitioners often get lost in technological space, pursuing the most advanced machine learning methods, the newest open source software libraries, or the glitziest visualization techniques. However, the heart of data science lies in doing the simple things right: understanding the application domain, cleaning and integrating relevant data sources, and presenting your results clearly to others.
  Simple doesn't mean easy, however. Indeed it takes considerable insight and experience to ask the right questions, and sense whether you are moving toward correct answers and actionable insights. I resist the temptation to drill deeply into clean, technical material here just because it is teachable. There are plenty of other books which will cover the intricacies of machine learning algorithms or statistical hypothesis testing. My mission here is to lay the groundwork of what really matters in analyzing data.

- *Developing mathematical intuition*: Data science rests on a foundation of mathematics, particularly statistics and linear algebra. It is important to understand this material on an intuitive level: why these concepts were developed, how they are useful, and when they work best. I illustrate operations in linear algebra by presenting pictures of what happens to matrices when you manipulate them, and statistical concepts by examples and *reducto ad absurdum* arguments. My goal here is transplanting intuition into the reader.
  But I strive to minimize the amount of formal mathematics used in presenting this material. Indeed, I will present exactly one formal proof in this book, an incorrect proof where the associated theorem is obviously false. The moral here is not that mathematical rigor doesn't matter, because of course it does, but that genuine rigor is impossible until after there is comprehension.

- *Think like a computer scientist, but act like a statistician*: Data science provides an umbrella linking computer scientists, statisticians, and domain specialists. But each community has its own distinct styles of thinking and action, which gets stamped into the souls of its members.
  In this book, I emphasize approaches which come most naturally to computer scientists, particularly the algorithmic manipulation of data, the use of machine learning, and the mastery of scale. But I also seek to transmit the core values of statistical reasoning: the need to understand the application domain, proper appreciation of the small, the quest for significance, and a hunger for exploration.
  No discipline has a monopoly on the truth. The best data scientists incor-

porate tools from multiple areas, and this book strives to be a relatively neutral ground where rival philosophies can come to reason together.

Equally important is what you will not find in this book. I do not emphasize any particular language or suite of data analysis tools. Instead, this book provides a high-level discussion of important design principles. I seek to operate at a conceptual level more than a technical one. The goal of this manual is to get you going in the right direction as quickly as possible, with whatever software tools you find most accessible.

**To the Instructor** This book covers enough material for an *"Introduction to Data Science"* course at the undergraduate or early graduate student levels. I hope that the reader has completed the equivalent of at least one programming course and has a bit of prior exposure to probability and statistics, but more is always better than less.

I have made a full set of lecture slides for teaching this course available online at http://www.data-manual.com. Data resources for projects and assignments are also available there to aid the instructor. Further, I make available online video lectures using these slides to teach a full-semester data science course. Let me help teach your class, through the magic of the web!

Pedagogical features of this book include:

- *War Stories*: To provide a better perspective on how data science techniques apply to the real world, I include a collection of "war stories," or tales from our experience with real problems. The moral of these stories is that these methods are not just theory, but important tools to be pulled out and used as needed.

- *False Starts*: Most textbooks present methods as a fait accompli, obscuring the ideas involved in designing them, and the subtle reasons why other approaches fail. The war stories illustrate my reasoning process on certain applied problems, but I weave such coverage into the core material as well.

- *Take-Home Lessons*: Highlighted "take-home" lesson boxes scattered through each chapter emphasize the big-picture concepts to learn from each chapter.

- *Homework Problems*: I provide a wide range of exercises for homework and self-study. Many are traditional exam-style problems, but there are also larger-scale implementation challenges and smaller-scale interview questions, reflecting the questions students might encounter when searching for a job. Degree of difficulty ratings have been assigned to all problems.

  In lieu of an answer key, a Solution Wiki has been set up, where solutions to all even numbered problems will be solicited by crowdsourcing. A similar system with my *Algorithm Design Manual* produced coherent solutions, or so I am told. As a matter of principle I refuse to look at them, so let the buyer beware.

- *Kaggle Challenges*: Kaggle (www.kaggle.com) provides a forum for data scientists to compete in, featuring challenging real-world problems on fascinating data sets, and scoring to test how good your model is relative to other submissions. The exercises for each chapter include three relevant Kaggle challenges, to serve as a source of inspiration, self-study, and data for other projects and investigations.

- *Data Science Television*: Data science remains mysterious and even threatening to the broader public. The *Quant Shop* is an amateur take on what a data science reality show should be like. Student teams tackle a diverse array of real-world prediction problems, and try to forecast the outcome of future events. Check it out at http://www.quant-shop.com.
  A series of eight 30-minute episodes has been prepared, each built around a particular real-world prediction problem. Challenges include pricing art at an auction, picking the winner of the Miss Universe competition, and forecasting when celebrities are destined to die. For each, we observe as a student team comes to grips with the problem, and learn along with them as they build a forecasting model. They make their predictions, and we watch along with them to see if they are right or wrong.
  In this book, *The Quant Shop* is used to provide concrete examples of prediction challenges, to frame discussions of the data science modeling pipeline from data acquisition to evaluation. I hope you find them fun, and that they will encourage you to conceive and take on your own modeling challenges.

- *Chapter Notes*: Finally, each tutorial chapter concludes with a brief notes section, pointing readers to primary sources and additional references.

**Dedication** My bright and loving daughters Bonnie and Abby are now full-blown teenagers, meaning that they don't always process statistical evidence with as much alacrity as I would I desire. I dedicate this book to them, in the hope that their analysis skills improve to the point that they always just agree with me.

And I dedicate this book to my beautiful wife Renee, who agrees with me even when she doesn't agree with me, and loves me beyond the support of all creditable evidence.

**Acknowledgments** My list of people to thank is large enough that I have probably missed some. I will try to do enumerate them systematically to minimize omissions, but ask those I've unfairly neglected for absolution.

First, I thank those who made concrete contributions to help me put this book together. Yeseul Lee served as an apprentice on this project, helping with figures, exercises, and more during summer 2016 and beyond. You will see evidence of her handiwork on almost every page, and I greatly appreciate her help and dedication. Aakriti Mittal and Jack Zheng also contributed to a few of the figures.

Students in my Fall 2016 *Introduction to Data Science* course (CSE 519) helped to debug the manuscript, and they found plenty of things to debug. I particularly thank Rebecca Siford, who proposed over one hundred corrections on her own. Several data science friends/sages reviewed specific chapters for me, and I thank Anshul Gandhi, Yifan Hu, Klaus Mueller, Francesco Orabona, Andy Schwartz, and Charles Ward for their efforts here.

I thank all the Quant Shop students from Fall 2015 whose video and modeling efforts are so visibly on display. I particularly thank Jan (Dini) DiskinZimmerman, whose editing efforts went so far beyond the call of duty I felt like a felon for letting her do it.

My editors at Springer, Wayne Wheeler and Simon Rees, were a pleasure to work with as usual. I also thank all the production and marketing people who helped get this book to you, including Adrian Pieron and Annette Anlauf.

Several exercises were originated by colleagues or inspired by other sources. Reconstructing the original sources years later can be challenging, but credits for each problem (to the best of my recollection) appear on the website.

Much of what I know about data science has been learned through working with other people. These include my Ph.D. students, particularly Rami al-Rfou, Mikhail Bautin, Haochen Chen, Yanqing Chen, Vivek Kulkarni, Levon Lloyd, Andrew Mehler, Bryan Perozzi, Yingtao Tian, Junting Ye, Wenbin Zhang, and postdoc Charles Ward. I fondly remember all of my Lydia project masters students over the years, and remind you that my prize offer to the first one who names their daughter Lydia remains unclaimed. I thank my other collaborators with stories to tell, including Bruce Futcher, Justin Gardin, Arnout van de Rijt, and Oleksii Starov.

I remember all members of the General Sentiment/Canrock universe, particularly Mark Fasciano, with whom I shared the start-up dream and experienced what happens when data hits the real world. I thank my colleagues at Yahoo Labs/Research during my 2015-2016 sabbatical year, when much of this book was conceived. I single out Amanda Stent, who enabled me to be at Yahoo during that particularly difficult year in the company's history. I learned valuable things from other people who have taught related data science courses, including Andrew Ng and Hans-Peter Pfister, and thank them all for their help.

If you have a procedure with ten parameters, you probably missed some.

- Alan Perlis

Caveat It is traditional for the author to magnanimously accept the blame for whatever deficiencies remain. I don't. Any errors, deficiencies, or problems in this book are somebody else's fault, but I would appreciate knowing about them so as to determine who is to blame.

# Chapter 2

# Contents

Chapter 1

# Chapter 3

# What is Data Science?

What is data science? Like any emerging field, it hasn't been completely defined yet, but you know enough about it to be interested or else you wouldn't be reading this book.

I think of data science as lying at the intersection of computer science, statistics, and substantive application domains. From computer science comes machine learning and high-performance computing technologies for dealing with scale. From statistics comes a long tradition of exploratory data analysis, significance testing, and visualization. From application domains in business and the sciences comes challenges worthy of battle, and evaluation standards to assess when they have been adequately conquered.

But these are all well-established fields. Why data science, and why now? I see three reasons for this sudden burst of activity:

- New technology makes it possible to capture, annotate, and store vast amounts of social media, logging, and sensor data. After you have amassed all this data, you begin to wonder what you can do with it.
- Computing advances make it possible to analyze data in novel ways and at ever increasing scales. Cloud computing architectures give even the little guy access to vast power when they need it. New approaches to machine learning have lead to amazing advances in longstanding problems, like computer vision and natural language processing.
- Prominent technology companies (like Google and Facebook) and quantitative hedge funds (like Renaissance Technologies and TwoSigma) have proven the power of modern data analytics. Success stories applying data to such diverse areas as sports management (Moneyball [ citelewis2004moneyball) and election forecasting (Nate Silver [ citesilver2012signal]) have served as role models to bring data science to a large popular audience.

This introductory chapter has three missions. First, I will try to explain how good data scientists think, and how this differs from the mindset of traditional programmers and software developers. Second, we will look at data sets in terms of the potential for what they can be used for, and learn to ask the broader questions they are capable of answering. Finally, I introduce a collection of data analysis challenges that will be used throughout this book as motivating examples.

Computer Science, Data Science, and Real Science Computer scientists, by nature, don't respect data. They have traditionally been taught that the algorithm was the thing, and that data was just meat to be passed through a sausage grinder.

So to qualify as an effective data scientist, you must first learn to think like a real scientist. Real scientists strive to understand the natural world, which is a complicated and messy place. By contrast, computer scientists tend to build their own clean and organized virtual worlds and live comfortably within them. Scientists obsess about discovering things, while computer scientists invent rather than discover.

People's mindsets strongly color how they think and act, causing misunderstandings when we try to communicate outside our tribes. So fundamental are these biases that we are often unaware we have them. Examples of the cultural differences between computer science and real science include:

- *Data vs. method centrism*: Scientists are data driven, while computer scientists are algorithm driven. Real scientists spend enormous amounts of effort collecting data to answer their question of interest. They invent fancy measuring devices, stay up all night tending to experiments, and devote most of their thinking to how to get the data they need.
  By contrast, computer scientists obsess about methods: which algorithm is better than which other algorithm, which programming language is best for a job, which program is better than which other program. The details of the data set they are working on seem comparably unexciting.

- *Concern about results*: Real scientists care about answers. They analyze data to discover something about how the world works. Good scientists care about whether the results make sense, because they care about what the answers mean.
  By contrast, bad computer scientists worry about producing plausiblelooking numbers. As soon as the numbers stop looking grossly wrong, they are presumed to be right. This is because they are personally less invested in what can be learned from a computation, as opposed to getting it done quickly and efficiently.

- *Robustness*: Real scientists are comfortable with the idea that data has errors. In general, computer scientists are not. Scientists think a lot about possible sources of bias or error in their data, and how these possible problems can effect the conclusions derived from them. Good programmers use strong data-typing and parsing methodologies to guard against formatting errors, but the concerns here are different.

Becoming aware that data can have errors is empowering. Computer scientists chant "garbage in, garbage out" as a defensive mantra to ward off criticism, a way to say *that's not my job*. Real scientists get close enough to their data to smell it, giving it the sniff test to decide whether it is likely to be garbage.

- *Precision*: Nothing is ever completely true or false in science, while *everything* is either true or false in computer science or mathematics. Generally speaking, computer scientists are happy printing floating point numbers to as many digits as possible: $8/13 = 0.61538461538$. Real scientists will use only two significant digits: $8/13 \approx 0.62$. Computer scientists care what a number is, while real scientists care what it means.

Aspiring data scientists must learn to think like real scientists. Your job is going to be to turn numbers into insight. It is important to understand the why as much as the how.

To be fair, it benefits real scientists to think like data scientists as well. New experimental technologies enable measuring systems on vastly greater scale than ever possible before, through technologies like full-genome sequencing in biology and full-sky telescope surveys in astronomy. With new breadth of view comes new levels of vision.

Traditional *hypothesis-driven* science was based on asking specific questions of the world and then generating the specific data needed to confirm or deny it. This is now augmented by *data-driven* science, which instead focuses on generating data on a previously unheard of scale or resolution, in the belief that new discoveries will come as soon as one is able to look at it. Both ways of thinking will be important to us:

- Given a problem, what available data will help us answer it?
- Given a data set, what interesting problems can we apply it to?

There is another way to capture this basic distinction between software engineering and data science. It is that software developers are hired to build systems, while data scientists are hired to produce insights.

This may be a point of contention for some developers. There exist an important class of engineers who wrangle the massive distributed infrastructures necessary to store and analyze, say, financial transaction or social media data on a full Facebook or Twitter-level of scale. Indeed, I will devote Chapter 12 to the distinctive challenges of big data infrastructures. These engineers are building tools and systems to support data science, even though they may not personally mine the data they wrangle. Do they qualify as data scientists?

This is a fair question, one I will finesse a bit so as to maximize the potential readership of this book. But I do believe that the better such engineers understand the full data analysis pipeline, the more likely they will be able to build powerful tools capable of providing important insights. A major goal of this book is providing big data engineers with the intellectual tools to think like big data scientists.

## 3.1 Asking Interesting Questions from Data

Good data scientists develop an inherent curiosity about the world around them, particularly in the associated domains and applications they are working on. They enjoy talking shop with the people whose data they work with. They ask them questions: What is the coolest thing you have learned about this field? Why did you get interested in it? What do you hope to learn by analyzing your data set? Data scientists always ask questions.

Good data scientists have wide-ranging interests. They read the newspaper every day to get a broader perspective on what is exciting. They understand that the world is an interesting place. Knowing a little something about everything equips them to play in other people's backyards. They are brave enough to get out of their comfort zones a bit, and driven to learn more once they get there.

Software developers are not really encouraged to ask questions, but data scientists are. We ask questions like:

- What things might you be able to learn from a given data set?
- What do you/your people really want to know about the world?
- What will it mean to you once you find out?

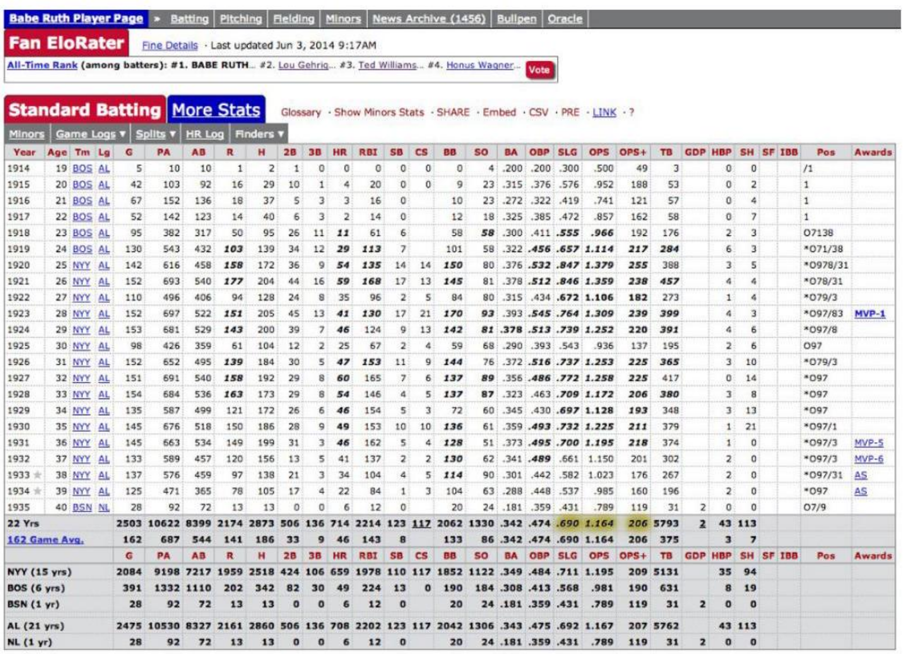
Computer scientists traditionally do not really appreciate data. Think about the way algorithm performance is experimentally measured. Usually the program is run on "random data" to see how long it takes. They rarely even look at the results of the computation, except to verify that it is correct and efficient. Since the "data" is meaningless, the results cannot be important. In contrast, real data sets are a scarce resource, which required hard work and imagination to obtain.

Becoming a data scientist requires learning to ask questions about data, so let's practice. Each of the subsections below will introduce an interesting data set. After you understand what kind of information is available, try to come up with, say, five interesting questions you might explore/answer with access to this data set.

The key is thinking broadly: the answers to big, general questions often lie buried in highly-specific data sets, which were by no means designed to contain them.

### 3.1.1 The Baseball Encyclopedia

Baseball has long had an outsized importance in the world of data science. This sport has been called the national pastime of the United States; indeed, French



Babe Ruth Player Page » Batting | Pitching | Fielding | Minors | News Archive (1456) | Bullpen | Oracle

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All-Time Rank (among batters): #1. BABE RUTH... #2. Lou Gehrig... #3. Ted Williams... #4. Honus Wagner... Vote

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| Year | Age | Tm | Lg | G | PA | AB | R | H | 2B | 3B | HR | RBI | SB | CS | BB | SO | BA | OBP | SLG | OPS | OPS+ | TB | GDP | HBP | SH | SF | IBB | Pos | Awards |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1914 | 19 | BOS | AL | 5 | 10 | 10 | 1 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | .200 | .200 | .300 | .500 | 49 | 3 | | 0 | 0 | | | /1 | |
| 1915 | 20 | BOS | AL | 42 | 103 | 92 | 16 | 29 | 10 | 1 | 4 | 20 | 0 | 0 | 9 | 23 | .315 | .376 | .576 | .952 | 188 | 53 | | 0 | 2 | | | 1 | |
| 1916 | 21 | BOS | AL | 67 | 152 | 136 | 18 | 37 | 5 | 3 | 3 | 16 | 0 | | 10 | 23 | .272 | .322 | .419 | .741 | 121 | 57 | | 0 | 4 | | | 1 | |
| 1917 | 22 | BOS | AL | 52 | 142 | 123 | 14 | 40 | 6 | 3 | 2 | 14 | 0 | | 12 | 18 | .325 | .385 | .472 | .857 | 162 | 58 | | 0 | 7 | | | 1 | |
| 1918 | 23 | BOS | AL | 95 | 382 | 317 | 50 | 95 | 26 | 11 | ***11*** | 61 | 6 | | 58 | ***58*** | .300 | .411 | ***.555*** | ***.966*** | 192 | 176 | | 2 | 3 | | | O7138 | |
| 1919 | 24 | BOS | AL | 130 | 543 | 432 | ***103*** | 139 | 34 | 12 | ***29*** | ***113*** | 7 | | 101 | 58 | .322 | ***.456*** | ***.657*** | ***1.114*** | ***217*** | ***284*** | | 6 | 3 | | | *O71/38 | |
| 1920 | 25 | NYY | AL | 142 | 616 | 458 | ***158*** | 172 | 36 | 9 | ***54*** | ***135*** | 14 | 14 | ***150*** | 80 | .376 | ***.532*** | ***.847*** | ***1.379*** | ***255*** | 388 | | 3 | 5 | | | *O978/31 | |
| 1921 | 26 | NYY | AL | 152 | 693 | 540 | ***177*** | 204 | 44 | 16 | ***59*** | ***168*** | 17 | 13 | ***145*** | 81 | .378 | ***.512*** | ***.846*** | ***1.359*** | ***238*** | ***457*** | | 4 | 4 | | | *O78/31 | |
| 1922 | 27 | NYY | AL | 110 | 496 | 406 | 94 | 128 | 24 | 8 | 35 | 96 | 2 | 5 | 84 | 80 | .315 | .434 | ***.672*** | ***1.106*** | ***182*** | 273 | | 1 | 4 | | | *O79/3 | |
| 1923 | 28 | NYY | AL | 152 | 697 | 522 | ***151*** | 205 | 45 | 13 | ***41*** | ***130*** | 17 | 21 | ***170*** | ***93*** | .393 | ***.545*** | ***.764*** | ***1.309*** | ***239*** | ***399*** | | 4 | 3 | | | *O97/83 | MVP-1 |
| 1924 | 29 | NYY | AL | 153 | 681 | 529 | ***143*** | 200 | 39 | 7 | ***46*** | 124 | 9 | 13 | ***142*** | ***81*** | ***.378*** | ***.513*** | ***.739*** | ***1.252*** | ***220*** | ***391*** | | 4 | 6 | | | *O97/8 | |
| 1925 | 30 | NYY | AL | 98 | 426 | 359 | 61 | 104 | 12 | 2 | 25 | 67 | 2 | 4 | 59 | 68 | .290 | .393 | .543 | .936 | 137 | 195 | | 2 | 6 | | | O97 | |
| 1926 | 31 | NYY | AL | 152 | 652 | 495 | ***139*** | 184 | 30 | 5 | ***47*** | ***153*** | 11 | 9 | ***144*** | 76 | .372 | ***.516*** | ***.737*** | ***1.253*** | ***225*** | ***365*** | | 3 | 10 | | | *O79/3 | |
| 1927 | 32 | NYY | AL | 151 | 691 | 540 | ***158*** | 192 | 29 | 8 | ***60*** | 165 | 7 | 6 | ***137*** | ***89*** | .356 | ***.486*** | ***.772*** | ***1.258*** | ***225*** | 417 | | 0 | 14 | | | *O97 | |
| 1928 | 33 | NYY | AL | 154 | 684 | 536 | ***163*** | 173 | 29 | 8 | ***54*** | 146 | 4 | 5 | ***137*** | ***87*** | .323 | .463 | ***.709*** | ***1.172*** | ***206*** | ***380*** | | 3 | 8 | | | *O97 | |
| 1929 | 34 | NYY | AL | 135 | 587 | 499 | 121 | 172 | 26 | 6 | ***46*** | 154 | 5 | 3 | 72 | 60 | .345 | .430 | ***.697*** | **1.128** | ***193*** | 348 | | 3 | 13 | | | *O97 | |
| 1930 | 35 | NYY | AL | 145 | 676 | 518 | 150 | 186 | 28 | 9 | **49** | 153 | 10 | 10 | ***136*** | 61 | .359 | ***.493*** | ***.732*** | ***1.225*** | ***211*** | 379 | | 1 | 21 | | | *O97/1 | |
| 1931 | 36 | NYY | AL | 145 | 663 | 534 | 149 | 199 | 31 | 3 | ***46*** | 162 | 5 | 4 | ***128*** | 51 | .373 | ***.495*** | ***.700*** | ***1.195*** | ***218*** | 374 | | 1 | 0 | | | *O97/3 | MVP-5 |
| 1932 | 37 | NYY | AL | 133 | 589 | 457 | 120 | 156 | 13 | 5 | 41 | 137 | 2 | 2 | ***130*** | 62 | .341 | ***.489*** | .661 | 1.150 | 201 | 302 | | 2 | 0 | | | *O97/3 | MVP-6 |
| 1933 ★ | 38 | NYY | AL | 137 | 576 | 459 | 97 | 138 | 21 | 3 | 34 | 104 | 4 | 5 | ***114*** | 90 | .301 | .442 | .582 | 1.023 | 176 | 267 | | 2 | 0 | | | *O97/31 | AS |
| 1934 ★ | 39 | NYY | AL | 125 | 471 | 365 | 78 | 105 | 17 | 4 | 22 | 84 | 1 | 3 | 104 | 63 | .288 | .448 | .537 | .985 | 160 | 196 | | 2 | 0 | | | *O97 | AS |
| 1935 | 40 | BSN | NL | 28 | 92 | 72 | 13 | 13 | 0 | 0 | 6 | 12 | 0 | | 20 | 24 | .181 | .359 | .431 | .789 | 119 | 31 | 2 | 0 | 0 | | | O7/9 | |
| **22 Yrs** | | | | **2503** | **10622** | **8399** | **2174** | **2873** | **506** | **136** | **714** | **2214** | **123** | **117** | **2062** | **1330** | **.342** | **.474** | ***.690*** | ***1.164*** | ***206*** | **5793** | **2** | **43** | **113** | | | | |
| **162 Game Avg.** | | | | **162** | **687** | **544** | **141** | **186** | **33** | **9** | **46** | **143** | **8** | | **133** | **86** | **.342** | **.474** | **.690** | **1.164** | **206** | **375** | | **3** | **7** | | | | |
| | | | | G | PA | AB | R | H | 2B | 3B | HR | RBI | SB | CS | BB | SO | BA | OBP | SLG | OPS | OPS+ | TB | GDP | HBP | SH | SF | IBB | Pos | Awards |
| **NYY (15 yrs)** | | | | **2084** | **9198** | **7217** | **1959** | **2518** | **424** | **106** | **659** | **1978** | **110** | **117** | **1852** | **1122** | **.349** | **.484** | **.711** | **1.195** | **209** | **5131** | | **35** | **94** | | | | |
| **BOS (6 yrs)** | | | | **391** | **1332** | **1110** | **202** | **342** | **82** | **30** | **49** | **224** | **13** | **0** | **190** | **184** | **.308** | **.413** | **.568** | **.981** | **190** | **631** | | **8** | **19** | | | | |
| **BSN (1 yr)** | | | | **28** | **92** | **72** | **13** | **13** | **0** | **0** | **6** | **12** | **0** | | **20** | **24** | **.181** | **.359** | **.431** | **.789** | **119** | **31** | **2** | **0** | **0** | | | | |
| **AL (21 yrs)** | | | | **2475** | **10530** | **8327** | **2161** | **2860** | **506** | **136** | **708** | **2202** | **123** | **117** | **2042** | **1306** | **.343** | **.475** | **.692** | **1.167** | **207** | **5762** | | **43** | **113** | | | | |
| **NL (1 yr)** | | | | **28** | **92** | **72** | **13** | **13** | **0** | **0** | **6** | **12** | **0** | | **20** | **24** | **.181** | **.359** | **.431** | **.789** | **119** | **31** | **2** | **0** | **0** | | | | |

Figure 3.1: //www.baseball-reference.com
http://www.baseball-reference.com.

historian Jacques Barzun observed that "Whoever wants to know the heart and mind of America had better learn baseball." I realize that many readers are not American, and even those that are might be completely disinterested in sports. But stick with me for a while.

What makes baseball important to data science is its extensive statistical record of play, dating back for well over a hundred years. Baseball is a sport of discrete events: pitchers throw balls and batters try to hit them - that naturally lends itself to informative statistics. Fans get immersed in these statistics as children, building their intuition about the strengths and limitations of quantitative analysis. Some of these children grow up to become data scientists. Indeed, the success of Brad Pitt's statistically-minded baseball team in the movie *Moneyball* remains the American public's most vivid contact with data science.

This historical baseball record is available at http://www. baseball-reference . com. There you will find complete statistical data on the performance of every player who even stepped on the field. This includes summary statistics of each season's batting, pitching, and fielding record, plus information about teams

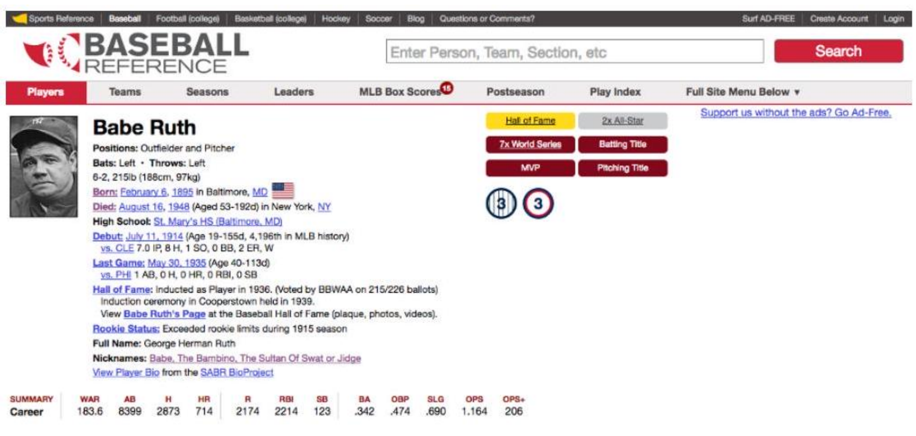


Figure 3.2: //www.baseball-reference.com
http://www.baseball-reference.com.

and awards as shown in Figure 1.1
But more than just statistics, there is metadata on the life and careers of all the people who have ever played major league baseball, as shown in Figure 1.2 We get the vital statistics of each player (height, weight, handedness) and their lifespan (when/where they were born and died). We also get salary information (how much each player got paid every season) and transaction data (how did they get to be the property of each team they played for).

Now, I realize that many of you do not have the slightest knowledge of or interest in baseball. This sport is somewhat reminiscent of cricket, if that helps. But remember that as a data scientist, it is your job to be interested in the world around you. Think of this as chance to learn something.

So what interesting questions can you answer with this baseball data set? Try to write down five questions before moving on. Don't worry, I will wait here for you to finish.

The most obvious types of questions to answer with this data are directly related to baseball:

- How can we best measure an individual player's skill or value?
- How fairly do trades between teams generally work out?
- What is the general trajectory of player's performance level as they mature and age?
- To what extent does batting performance correlate with position played? For example, are outfielders really better hitters than infielders?

These are interesting questions. But even more interesting are questions about demographic and social issues. Almost 20,000 major league baseball play-

ers have taken the field over the past 150 years, providing a large, extensivelydocumented cohort of men who can serve as a proxy for even larger, less welldocumented populations. Indeed, we can use this baseball player data to answer questions like:

- Do left-handed people have shorter lifespans than right-handers? Handedness is not captured in most demographic data sets, but has been diligently assembled here. Indeed, analysis of this data set has been used to show that right-handed people live longer than lefties [ citehalpern1988do]
- How often do people return to live in the same place where they were born? Locations of birth and death have been extensively recorded in this data set. Further, almost all of these people played at least part of their career far from home, thus exposing them to the wider world at a critical time in their youth.
- Do player salaries generally reflect past, present, or future performance?
- To what extent have heights and weights been increasing in the population at large?

There are two particular themes to be aware of here. First, the identifiers and reference tags (i.e. the metadata) often prove more interesting in a data set than the stuff we are supposed to care about, here the statistical record of play.

Second is the idea of a *statistical proxy*, where you use the data set you have to substitute for the one you really want. The data set of your dreams likely does not exist, or may be locked away behind a corporate wall even if it does. A good data scientist is a pragmatist, seeing what they can do with what they have instead of bemoaning what they cannot get their hands on.

### 3.1.2 The Internet Movie Database (IMDb)

Everybody loves the movies. The Internet Movie Database (IMDb) provides crowdsourced and curated data about all aspects of the motion picture industry, at www.imdb.com. IMDb currently contains data on over 3.3 million movies and TV programs. For each film, IMDb includes its title, running time, genres, date of release, and a full list of cast and crew. There is financial data about each production, including the budget for making the film and how well it did at the box office.

Finally, there are extensive ratings for each film from viewers and critics. This rating data consists of scores on a zero to ten stars scale, cross-tabulated into averages by age and gender. Written reviews are often included, explaining why a particular critic awarded a given number of stars. There are also links between films: for example, identifying which other films have been watched most often by viewers of *It's a Wonderful Life*.

Every actor, director, producer, and crew member associated with a film merits an entry in IMDb, which now contains records on 6.5 million people.


Figure 3.3: Representative film data from the Internet Movie Database.

Won 1 Oscar. Another 25 wins & 19 nominations. See more awards *
Figure 1.4: Representative actor data from the Internet Movie Database.


Figure 3.4: Representative film data from the Internet Movie Database.

These happen to include my brother, cousin, and sister-in-law. Each actor is linked to every film *they* appeared in, with a description of their role and their ordering in the credits. Available *data* about each personality includes birth/death dates, height, awards, and family relations.

So what kind of questions can you answer with this movie data?

Perhaps the most natural questions to ask IMDb involve identifying the extremes of movies and actors:

- Which actors appeared in the most films? Earned the most money? Appeared in the lowest rated films? Had the longest career or the shortest lifespan?
- What was the highest rated film each year, or the best in each genre? Which movies lost the most money, had the highest-powered casts, or got the least favorable reviews.

Then there are larger-scale questions one can ask about the nature of the motion picture business itself:

- How well does movie gross correlate with viewer ratings or awards? Do customers instinctively flock to trash, or is virtue on the part of the creative team properly rewarded?
- How do Hollywood movies compare to Bollywood movies, in terms of ratings, budget, and gross? Are American movies better received than foreign films, and how does this differ between U.S. and non-U.S. reviewers?
- What is the age distribution of actors and actresses in films? How much younger is the actress playing the wife, on average, than the actor playing the husband? Has this disparity been increasing or decreasing with time?
- Live fast, die young, and leave a good-looking corpse? Do movie stars live longer or shorter lives than bit players, or compared to the general public?

Assuming that people working together on a film get to know each other, the cast and crew data can be used to build a social network of the movie business. What does the social network of actors look like? The Oracle of Bacon (https://oracleofbacon.org/) posits Kevin Bacon as the center of the Hollywood universe and generates the shortest path to Bacon from any other actor. Other actors, like Samuel L. Jackson, prove even more central.

More critically, can we analyze this data to determine the probability that someone will like a given movie? The technique of *collaborative filtering* finds people who liked films that I also liked, and recommends other films that they liked as good candidates for me. The 2007 Netflix Prize was a $\$1,000,000$ competition to produce a ratings engine $10\%$ better than the proprietary Netflix system. The ultimate winner of this prize (BellKor) used a variety of data sources and techniques, including the analysis of links [
citebell2007lessons.

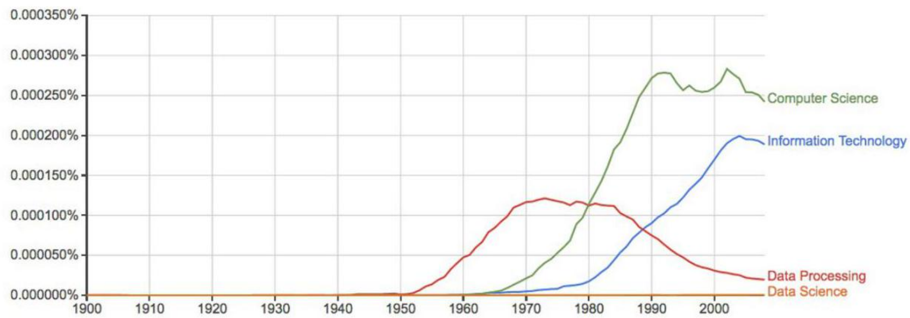


Figure 3.5: The rise and fall of data processing, as witnessed by *Google Ngrams*.

### 3.1.3 Google Ngrams

Printed books have been the primary repository of human knowledge since Gutenberg's invention of movable type in 1439. Physical objects live somewhat uneasily in today's digital world, but technology has a way of reducing everything to data. As part of its mission to organize the world's information, Google undertook an effort to scan all of the world's published books. They haven't quite gotten there yet, but the 30 million books thus far digitized represent over 20% of all books ever published.

Google uses this data to improve search results, and provide fresh access to out-of-print books. But perhaps the coolest product is Google Ngrams, an amazing resource for monitoring changes in the cultural zeitgeist. It provides the frequency with which short phrases occur in books published each year. Each phrase must occur at least forty times in their scanned book corpus. This eliminates obscure words and phrases, but leaves over two billion time series available for analysis.

This rich data set shows how language use has changed over the past 200 years, and has been widely applied to cultural trend analysis [MAV $^{+}$11. Figure 1.5 uses this data to show how the word data fell out of favor when thinking about computing. *Data processing* was the popular term associated with the computing field during the punched card and spinning magnetic tape era of the 1950s. The Ngrams data shows that the rapid rise of *Computer Science* did not eclipse *Data Processing* until 1980. Even today, *Data Science* remains almost invisible on this scale.

Check out Google Ngrams at http://books.google.com/ngrams. I promise you will enjoy playing with it. Compare *hot dog* to *tofu*, *science* against *religion*, *freedom* to *justice*, and sex vs. *marriage*, to better understand this fantastic telescope for looking into the past.

But once you are done playing, think of bigger things you could do if you got your hands on this data. Assume you have access to the annual number of references for all words/phrases published in books over the past 200 years.

Google makes this data freely available. So what are you going to do with it?

Observing the time series associated with particular words using the Ngrams Viewer is fun. But more sophisticated historical trends can be captured by aggregating multiple time series together. The following types of questions seem particularly interesting to me:

- How has the amount of cursing changed over time? Use of the fourletter words I am most familiar with seem to have exploded since 1960, although it is perhaps less clear whether this reflects increased cussing or lower publication standards.

- How often do new words emerge and get popular? Do these words tend to stay in common usage, or rapidly fade away? Can we detect when words change meaning over time, like the transition of gay from *happy* to *homosexual*?

- Have standards of spelling been improving or deteriorating with time, especially now that we have entered the era of automated spell checking? Rarely-occurring words that are only one character removed from a commonly-used word are likely candidates to be spelling errors (e.g. *algorithm* vs. *algorthm*). Aggregated over many different misspellings, are such errors increasing or decreasing?

You can also use this Ngrams corpus to build a language model that captures the meaning and usage of the words in a given language. We will discuss word embeddings in Section 11.6 .3 which are powerful tools for building language models. Frequency counts reveal which words are most popular. The frequency of word pairs appearing next to each other can be used to improve speech recognition systems, helping to distinguish whether the speaker said that's too bad or *that's to bad*. These millions of books provide an ample data set to build representative models from.

### 3.1.4 New York Taxi Records

Every financial transaction today leaves a data trail behind it. Following these paths can lead to interesting insights.

Taxi cabs form an important part of the urban transportation network. They roam the streets of the city looking for customers, and then drive them to their destination for a fare proportional to the length of the trip. Each cab contains a metering device to calculate the cost of the trip as a function of time. This meter serves as a record keeping device, and a mechanism to ensure that the driver charges the proper amount for each trip.

The taxi meters currently employed in New York cabs can do many things beyond calculating fares. They act as credit card terminals, providing a way

| Vendor ID | passenger count | trip distance | pickup longitude | pickup_ latitude height2 | 1 | 7.22 |
|---|---|---|---|---|---|---|
| 40.80662 | 2 | 0 | 30.8 | | | |
| 1 | 1 | 2.3 | -73.977 | 40.7749 | -73.9783 | 40.74986 |
| 1 | 1 | 1.5 | -73.9591 | 40.77513 | -73.9804 | 40.78231 |
| 1 | 1 | 0.9 | -73.9766 | 40.78075 | -73.9706 | 40.78885 |
| 2 | 1 | 2.44 | -73.9786 | 40.78592 | -73.9974 | 40.7563 |
| 2 | 1 | 3.36 | -73.9764 | 40.78589 | -73.9424 | 40.82209 |
| 2 | 2 | 2.34 | -73.9862 | 40.76087 | -73.9569 | 40.77156 |
| 2 | 1 | 10.19 | -73.79 | 40.64406 | -73.9312 | 40.67588 |
| 1 | 2 | 3.3 | -73.9937 | 40.72738 | -73.9982 | 40.7641 |
| 1 | 1 | 1.8 | -73.9949 | 40.74006 | -73.9767 | 40.74934 |

Figure 1.6: Representative fields from the New York city taxi cab data: pick up and dropoff points, distances, and fares.

for customers to pay for rides without cash. They are integrated with global positioning systems (GPS), recording the exact location of every pickup and drop off. And finally, since they are on a wireless network, these boxes can communicate all of this data back to a central server.

The result is a database documenting every single trip by all taxi cabs in one of the world's greatest cities, a small portion of which is shown in Figure 1.6. Because the New York Taxi and Limousine Commission is a public agency, its non-confidential data is available to all under the Freedom of Information Act (FOA).

Every ride generates two records: one with data on the trip, the other with details of the fare. Each trip is keyed to the medallion (license) of each car coupled with the identifier of each driver. For each trip, we get the time/date of pickup and drop-off, as well as the GPS coordinates (longitude and latitude) of the starting location and destination. We do not get GPS data of the route they traveled between these points, but to some extent that can be inferred by the shortest path between them.

As for fare data, we get the metered cost of each trip, including tax, surcharge and tolls. It is traditional to pay the driver a tip for service, the amount of which is also recorded in the data.

So I'm talking to you. This taxi data is readily available, with records of over 80 million trips over the past several years. What are you going to do with it?

Any interesting data set can be used to answer questions on many different scales. This taxi fare data can help us better understand the transportation industry, but also how the city works and how we could make it work even better. Natural questions with respect to the taxi industry include:

- How much money do drivers make each night, on average? What is the distribution? Do drivers make more on sunny days or rainy days?
- Where are the best spots in the city for drivers to cruise, in order to pick up profitable fares? How does this vary at different times of the day?

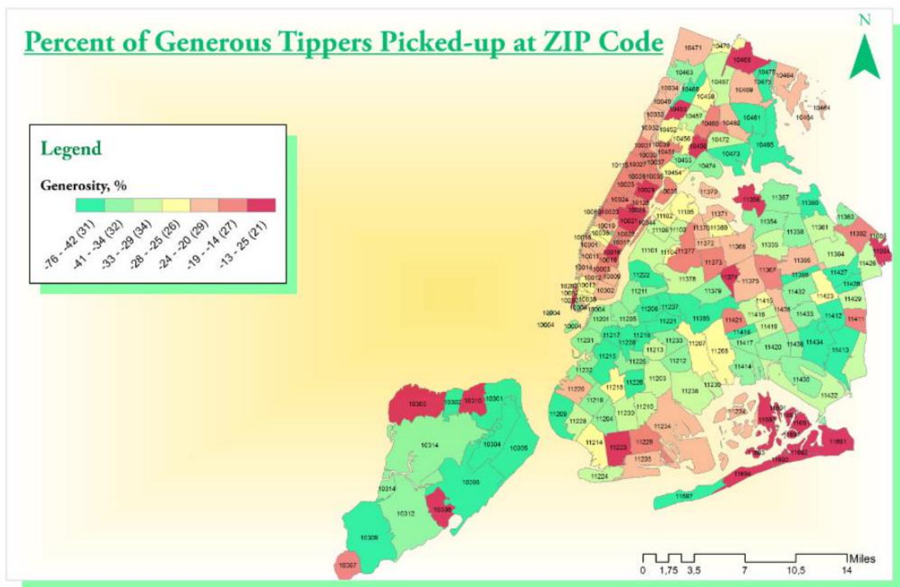


Figure 3.6: Which neighborhoods in New York city tip most generously? The relatively remote outer boroughs of Brooklyn and Queens, where trips are longest and supply is relatively scarce.

- How far do drivers travel over the course of a night's work? We can't answer this exactly using this data set, because it does not provide GPS data of the route traveled between fares. But we do know the last place of drop off, the next place of pickup, and how long it took to get between them. Together, this should provide enough information to make a sound estimate.

- Which drivers take their unsuspecting out-of-town passengers for a "ride," running up the meter on what should be a much shorter, cheaper trip?

- How much are drivers tipped, and why? Do faster drivers get tipped better? How do tipping rates vary by neighborhood, and is it the rich neighborhoods or poor neighborhoods which prove more generous?
  I will confess we did an analysis of this, which I will further describe in the war story of Section 9.3 . We found a variety of interesting patterns citestarov2015gis. Figure 1.7 shows that Manhattanites are generally cheapskates relative to large swaths of Brooklyn, Queens, and Staten Island, where trips are longer and street cabs a rare but welcome sight.

But the bigger questions have to do with understanding transportation in the city. We can use the taxi travel times as a sensor to measure the level of traffic in the city at a fine level. How much slower is traffic during rush hour than other times, and where are delays the worst? Identifying problem areas is the first step to proposing solutions, by changing the timing patterns of traffic lights, running more buses, or creating high-occupancy only lanes.

Similarly we can use the taxi data to measure transportation flows across the city. Where are people traveling to, at different times of the day? This tells us much more than just congestion. By looking at the taxi data, we should be able to see tourists going from hotels to attractions, executives from fancy neighborhoods to Wall Street, and drunks returning home from nightclubs after a bender.

Data like this is essential to designing better transportation systems. It is wasteful for a single rider to travel from point $a$ to point $b$ when there is another rider at point $a + \epsilon$ who also wants to get there. Analysis of the taxi data enables accurate simulation of a ride sharing system, so we can accurately evaluate the demands and cost reductions of such a service.

## 3.2 Properties of Data

This book is about techniques for analyzing data. But what is the underlying stuff that we will be studying? This section provides a brief taxonomy of the properties of data, so we can better appreciate and understand what we will be working on.

### 3.2.1 Structured vs. Unstructured Data

Certain data sets are nicely structured, like the tables in a database or spreadsheet program. Others record information about the state of the world, but in a more heterogeneous way. Perhaps it is a large text corpus with images and links like Wikipedia, or the complicated mix of notes and test results appearing in personal medical records.

Generally speaking, this book will focus on dealing with structured data. Data is often represented by a *matrix*, where the rows of the matrix represent distinct items or records, and the columns represent distinct properties of these items. For example, a data set about U.S. cities might contain one row for each city, with columns representing features like state, population, and area.

When confronted with an unstructured data source, such as a collection of tweets from Twitter, our first step is generally to build a matrix to structure it. A *bag of words* model will construct a matrix with a row for each tweet, and a column for each frequently used vocabulary word. Matrix entry $M[i, j]$ then denotes the number of times tweet $i$ contains word $j$. Such matrix formulations will motivate our discussion of linear algebra, in Chapter 8

### 3.2.2 Quantitative vs. Categorical Data

*Quantitative data* consists of numerical values, like height and weight. Such data can be incorporated directly into algebraic formulas and mathematical models, or displayed in conventional graphs and charts.

By contrast, *categorical data* consists of labels describing the properties of the objects under investigation, like gender, hair color, and occupation. This descriptive information can be every bit as precise and meaningful as numerical data, but it cannot be worked with using the same techniques.

Categorical data can usually be coded numerically. For example, gender might be represented as *male* $= 0$ or *female* $= 1$. But things get more complicated when there are more than two characters per feature, especially when there is not an implicit order between them. We may be able to encode hair colors as numbers by assigning each shade a distinct value like gray hair $= 0$, red hair $= 1$, and blond hair $= 2$. However, we cannot really treat these values as numbers, for anything other than simple identity testing. Does it make any sense to talk about the maximum or minimum hair color? What is the interpretation of my hair color minus your hair color?

Most of what we do in this book will revolve around numerical data. But keep an eye out for categorical features, and methods that work for them. *Classification* and clustering methods can be thought of as generating categorical labels from numerical data, and will be a primary focus in this book.

### 3.2.3 Big Data vs. Little Data

Data science has become conflated in the public eye with *big data*, the analysis of massive data sets resulting from computer logs and sensor devices. In principle, having more data is always better than having less, because you can always throw some of it away by sampling to get a smaller set if necessary.

*Big data* is an exciting phenomenon, and we will discuss it in Chapter 12 But in practice, there are difficulties in working with large data sets. Throughout this book we will look at algorithms and best practices for analyzing data. In general, things get harder once the volume gets too large. The challenges of big data include:

- *The analysis cycle time slows as data size grows*: Computational operations on data sets take longer as their volume increases. Small spreadsheets provide instantaneous response, allowing you to experiment and play what $if$ ? But large spreadsheets can be slow and clumsy to work with, and massive-enough data sets might take hours or days to get answers from.

Clever algorithms can permit amazing things to be done with big data, but staying small generally leads to faster analysis and exploration.

- *Large data sets are complex to visualize*: Plots with millions of points on them are impossible to display on computer screens or printed images, let alone conceptually understand. How can we ever hope to really understand something we cannot see?

- *Simple models do not require massive data to fit or evaluate*: A typical data science task might be to make a decision (say, whether I should offer this fellow life insurance?) on the basis of a small number of variables: say age, gender, height, weight, and the presence or absence of existing medical conditions.

If I have this data on 1 million people with their associated life outcomes, I should be able to build a good general model of coverage risk. It probably wouldn't help me build a substantially better model if I had this data on hundreds of millions of people. The decision criteria on only a few variables (like age and martial status) cannot be too complex, and should be robust over a large number of applicants. Any observation that is so subtle it requires massive data to tease out will prove irrelevant to a large business which is based on volume.

Big data is sometimes called *bad data*. It is often gathered as the by-product of a given system or procedure, instead of being purposefully collected to answer your question at hand. The result is that we might have to go to heroic efforts to make sense of something just because we have it.

Consider the problem of getting a pulse on voter preferences among presidential candidates. The big data approach might analyze massive Twitter or Facebook feeds, interpreting clues to their opinions in the text. The small data approach might be to conduct a poll, asking a few hundred people this specific question and tabulating the results. Which procedure do you think will prove more accurate? The *right* data set is the one most directly relevant to the tasks at hand, not necessarily the biggest one.

*Take-Home Lesson*: Do not blindly aspire to analyze large data sets. Seek the right data to answer a given question, not necessarily the biggest thing you can get your hands on.

Classification and *Regression* Two types of problems arise repeatedly in traditional data science and pattern recognition applications, the challenges of *classification* and *regression*. As this book has developed, I have pushed discussions of the algorithmic approaches to solving these problems toward the later chapters, so they can benefit from a solid understanding of core material in data munging, statistics, visualization, and mathematical modeling.

Still, I will mention issues related to classification and regression as they arise, so it makes sense to pause here for a quick introduction to these problems, to help you recognize them when you see them.

- Classification: Often we seek to assign a label to an item from a discrete set of possibilities. Such problems as predicting the winner of a particular sporting contest (team $A$ or team $B$ ?) or deciding the genre of a given movie (comedy, drama, or animation?) are classification problems, since each entail selecting a label from the possible choices.

- Regression: Another common task is to forecast a given numerical quantity. Predicting a person's weight or how much snow we will get this year is a regression problem, where we forecast the future value of a numerical function in terms of previous values and other relevant features.

Perhaps the best way to see the intended distinction is to look at a variety of data science problems and label (classify) them as regression or classification. Different algorithmic methods are used to solve these two types of problems, although the same questions can often be approached in either way:

- Will the price of a particular stock be higher or lower tomorrow? (classification)

- What will the price of a particular stock be tomorrow? (regression)

- Is this person a good risk to sell an insurance policy to? (classification)

- How long do we expect this person to live? (regression)

Keep your eyes open for classification and regression problems as you encounter them in your life, and in this book.

Data Science Television: *The Quant Shop* I believe that hands-on experience is necessary to internalize basic principles. Thus when I teach data science, I like to give each student team an interesting but messy forecasting challenge, and demand that they build and evaluate a predictive model for the task.

These forecasting challenges are associated with events where the students must make testable predictions. They start from scratch: finding the relevant data sets, building their own evaluation environments, and devising their model.

Finally, I make them watch the event as it unfolds, so as to witness the vindication or collapse of their prediction.

As an experiment, we documented the evolution of each group's project on video in Fall 2014. Professionally edited, this became The Quant Shop, a television-like data science series for a general audience. The eight episodes of this first season are available at http://www.quant-shop.com and include:

- *Finding Miss Universe* - The annual Miss Universe competition aspires to identify the most beautiful woman in the world. Can computational models predict who will win a beauty contest? Is beauty just subjective, or can algorithms tell who is the fairest one of all?

- *Modeling the Movies* - The business of movie making involves a lot of high-stakes data analysis. Can we build models to predict which film will gross the most on Christmas day? How about identifying which actors will receive awards for their performance?

- *Winning the Baby Pool* - Birth weight is an important factor in assessing the health of a newborn child. But how accurately can we predict junior's weight before the actual birth? How can data clarify environmental risks to developing pregnancies?

- *The Art of the Auction* - The world's most valuable artworks sell at auctions to the highest bidder. But can we predict how many millions a particular J.W. Turner painting will sell for? Can computers develop an artistic sense of what's worth buying?

- *White Christmas* - Weather forecasting is perhaps the most familiar domain of predictive modeling. Short-term forecasts are generally accurate, but what about longer-term prediction? What places will wake up to a snowy Christmas this year? And can you tell one month in advance?

- *Predicting the Playoffs* - Sports events have winners and losers, and bookies are happy to take your bets on the outcome of any match. How well can statistics help predict which football team will win the Super Bowl? Can Google's PageRank algorithm pick the winners on the field as accurately as it does on the web?

- *The Ghoul Pool* - Death comes to all men, but when? Can we apply actuarial models to celebrities, to decide who will be the next to die? Similar analysis underlies the workings of the life insurance industry, where accurate predictions of lifespan are necessary to set premiums which are both sustainable and affordable.

- *Playing the Market* - Hedge fund quants get rich when guessing right about tomorrow's prices, and poor when wrong. How accurately can we predict future prices of gold and oil using histories of price data? What other information goes into building a successful price model?

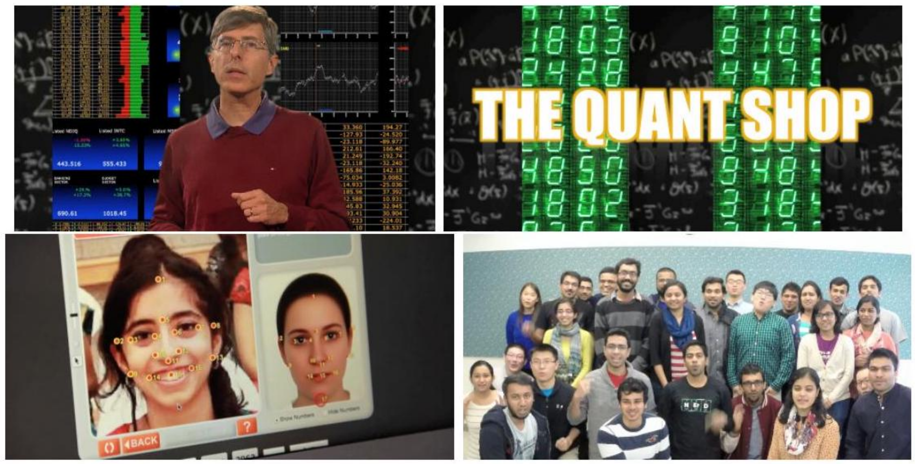


Figure 3.7: *The Quant Shop.*

I encourage you to watch some episodes of The Quant Shop in tandem with reading this book. We try to make it fun, although I am sure you will find plenty of things to cringe at. Each show runs for thirty minutes, and maybe will inspire you to tackle a prediction challenge of your own.

These programs will certainly give you more insight into these eight specific challenges. I will use these projects throughout this book to illustrate important lessons in how to do data science, both as positive and negative examples. These projects provide a laboratory to see how intelligent but inexperienced people not wildly unlike yourself thought about a data science problem, and what happened when they did.

### 3.2.4 Kaggle Challenges

Another source of inspiration are challenges from Kaggle (www.kaggle.com), which provides a competitive forum for data scientists. New challenges are posted on a regular basis, providing a problem definition, training data, and a scoring function over hidden evaluation data. A leader board displays the scores of the strongest competitors, so you can see how well your model stacks up in comparison with your opponents. The winners spill their modeling secrets during post-contest interviews, to help you improve your modeling skills.

Performing well on Kaggle challenges is an excellent credential to put on your resume to get a good job as a data scientist. Indeed, potential employers will track you down if you are a real Kaggle star. But the real reason to participate is that the problems are fun and inspiring, and practice helps make you a better data scientist.

The exercises at the end of each chapter point to expired Kaggle challenges, loosely connected to the material in that chapter. Be forewarned that Kaggle provides a misleading glamorous view of data science as applied machine learning, because it presents extremely well-defined problems with the hard work of data collection and cleaning already done for you. Still, I encourage you to check it out for inspiration, and as a source of data for new projects.

## 3.3 About the War Stories

*Genius* and wisdom are two distinct intellectual gifts. Genius shows in discovering the right answer, making imaginative mental leaps which overcome obstacles and challenges. *Wisdom* shows in avoiding obstacles in the first place, providing a sense of direction or guiding light that keeps us moving soundly in the right direction.

Genius is manifested in technical strength and depth, the ability to see things and do things that other people cannot. In contrast, wisdom comes from experience and general knowledge. It comes from listening to others. Wisdom comes from humility, observing how often you have been wrong in the past and figuring out why you were wrong, so as to better recognize future traps and avoid them.

Data science, like most things in life, benefits more from wisdom than from genius. In this book, I seek to pass on wisdom that I have accumulated the hard way through *war stories*, gleaned from a diverse set of projects I have worked on:

- *Large-scale text analytics and NLP*: My Data Science Laboratory at Stony Brook University works on a variety of projects in big data, including sentiment analysis from social media, historical trends analysis, deep learning approaches to natural language processing (NLP), and feature extraction from networks.

- *Start-up companies*: I served as co-founder and chief scientist to two data analytics companies: General Sentiment and Thrivemetrics. General Sentiment analyzed large-scale text streams from news, blogs, and social media to identify trends in the sentiment (positive or negative) associated with people, places, and things. Thrivemetrics applied this type of analysis to internal corporate communications, like email and messaging systems.
  Neither of these ventures left me wealthy enough to forgo my royalties from this book, but they did provide me with experience on cloud-based computing systems, and insight into how data is used in industry.

- *Collaborating with real scientists*: I have had several interesting collaborations with biologists and social scientists, which helped shape my understanding of the complexities of working with real data. Experimental data is horribly noisy and riddled with errors, yet you must do the best you can with what you have, in order to discover how the world works.

- *Building gambling systems*: A particularly amusing project was building a system to predict the results of jai-alai matches so we could bet on them, an experience recounted in my book *Calculated Bets: Computers, Gambling, and Mathematical Modeling to Win*
citeskiena2001calculated. Our system relied on web scraping for data collection, statistical analysis, simulation/modeling, and careful evaluation. We also have developed and evaluated predictive models for movie grosses [
citezhang2009improving], stock prices [
citezhang2010trading], and football games
citehong2010wisdom using social media analysis.

- *Ranking historical figures*: By analyzing Wikipedia to extract meaningful variables on over 800,000 historical figures, we developed a scoring function to rank them by their strength as historical memes. This ranking does a great job separating the greatest of the great (Jesus, Napoleon, Shakespeare, Mohammad, and Lincoln round out the top five) from lesser mortals, and served as the basis for our book *Who's Bigger?: Where Historical Figures Really Rank*
citeskiena2013whos.

All this experience drives what I teach in this book, especially the tales that I describe as war stories. *Every one of these war stories is true.* Of course, the stories improve somewhat in the retelling, and the dialogue has been punched up to make them more interesting to read. However, I have tried to honestly trace the process of going from a raw problem to a solution, so you can watch how it unfolded.

## 3.4 War Story: Answering the Right Question

Our research group at Stony Brook University developed an NLP-based system for analyzing millions of news, blogs and social media messages, and reducing this text to trends concerning all the entities under discussion. Counting the number of mentions each name receives in a text stream (volume) is easy, in principle. Determining whether the connotation of a particular reference is positive or negative (sentiment analysis) is hard. But our system did a pretty good job, particularly when aggregated over many references.

This technology served as the foundation for a social media analysis company named General Sentiment. It was exciting living through a start-up starting up, facing the challenges of raising money, hiring staff, and developing new products.

But perhaps the biggest problem we faced was answering the right question. The General Sentiment system recorded trends about the sentiment and volume for *every* person, place, and thing that was ever mentioned in news, blogs, and social media: over 20 million distinct entities. We monitored the reputations of celebrities and politicians. We monitored the fates of companies and products. We tracked the performance of sports teams, and the buzz about movies. We could do anything!

But it turns out that no one pays you to do anything. They pay you to do *something*, to solve a particular problem they have, or eliminate a specific pain point in their business. Being able to do anything proves to be a terrible sales strategy, because it requires you to find that need afresh for each and every customer.

Facebook didn't open up to the world until September 2006. So when General Sentiment started in 2008, we were at the very beginning of the social media era. We had lots of interest from major brands and advertising agencies which *knew* that social media was ready to explode. They knew this newfangled thing was important, and that they had to be there. They knew that proper analysis of social media data could give them fresh insights into what their customers were thinking. But they didn't know exactly what it was they really wanted to know.

One aircraft engine manufacturer was very interested in learning how much the kids talked about them on Facebook. We had to break it to them gently that the answer was zero. Other potential customers demanded proof that we were more accurate than the Nielsen television ratings. But of course, if you wanted Nielsen ratings then you should buy them from Nielsen. Our system provided different insights from a completely different world. But you had to know what you wanted in order to use them.

We did manage to get substantial contracts from a very diverse group of customers, including consumer brands like Toyota and Blackberry, governmental organizations like the Hawaii tourism office, and even the presidential campaign of Republican nominee Mitt Romney in 2012. Our analysts provided them insights into a wide variety of business issues:

- What did people think about Hawaii? (Answer: they think it is a very nice place to visit.)
- How quickly would Toyota's sentiment recover after news of serious brake problems in their cars? (Answer: about six months.)
- What did people think about Blackberry's new phone models? (Answer: they liked the iPhone much better.)
- How quickly would Romney's sentiment recover after insulting $47\%$ of the electorate in a recorded speech? (Answer: never.)

But each sale required entering a new universe, involving considerable effort and imagination on the part of our sales staff and research analysts. We never managed to get two customers in the same industry, which would have let us benefit from scale and accumulated wisdom.

Of course, the customer is always right. It was our fault that we could not explain to them the best way to use our technology. The lesson here is that the world will not beat a path to your door just for a new source of data. You must be able to supply the right questions before you can turn data into money.

Chapter Notes The idea of using historical records from baseball players to establish that lefthanders have shorter lifespans is due to Halpern and Coren citehalpern1988do,
citehalpern1991handedness, but their conclusion remains controversial. The percentage of left-handers in the population has been rapidly growing, and the observed effects may be a function of survivorship bias [
citemcmanus2004right]. So lefties, hang in there! Full disclosure: I am one of you.

The discipline of quantitative baseball analysis is sometimes called *sabermetrics*, and its leading light is a fellow named Bill James. I recommend budding data scientists read his *Historical Baseball Abstract*
citejames2010new as an excellent example of how one turns numbers into knowledge and understanding. *Time Magazine* once said of James: "Much of the joy of reading him comes from the extravagant spectacle of a first-rate mind wasting itself on baseball." I thank http://sports-reference.com for permission to use images of their website in this book. Ditto to Amazon, the owner of IMDb.

The potential of ride-sharing systems in New York was studied by Santi et. al. SRS $^{+}14$, who showed that almost $95\%$ of the trips could have been shared with no more than five minutes delay per trip.

The Lydia system for sentiment analysis is described in
citegodbole2007large. Methods to identify changes in word meaning through analysis of historical text corpora like Google Ngram are reported in
citekulkarni2015statistically.

## 3.5 Exercises

Identifying Data Sets 1-1. [3] Identify where interesting data sets relevant to the following domains can be found on the web:
(a) Books.
(b) Horse racing.
(c) Stock prices.
(d) Risks of diseases.
(e) Colleges and universities.
(f) Crime rates.
(g) Bird watching.

For each of these data sources, explain what you must do to turn this data into a usable format on your computer for analysis.
1-2. [3] Propose relevant data sources for the following *The Quant Shop* prediction challenges. Distinguish between sources of data that you are sure *somebody* must have, and those where the data is clearly available to you.
(a) *Miss Universe*.

(b) *Movie gross.*
(c) *Baby weight.*
(d) *Art auction price.*
(e) *White Christmas.*
(f) *Football champions.*
(g) *Ghoul pool.*
(h) *Gold/oil prices.*

1-3. [3] Visit http://data.gov, and identify five data sets that sound interesting to you. For each write a brief description, and propose three interesting things you might do with them.

Asking Questions 1-4. [3] For each of the following data sources, propose three interesting questions you can answer by analyzing them:
(a) Credit card billing data.

(b) Click data from http://www.Amazon.com
(c) White Pages residential/commercial telephone directory.

1-5. [5] Visit Entrez, the National Center for Biotechnology Information (NCBI) portal. Investigate what data sources are available, particularly the Pubmed and Genome resources. Propose three interesting projects to explore with each of them.

1-6. [5] You would like to conduct an experiment to establish whether your friends prefer the taste of regular Coke or Diet Coke. Briefly outline a design for such a study.
1-7. [5] You would like to conduct an experiment to see whether students learn better if they study without any music, with instrumental music, or with songs that have lyrics. Briefly outline the design for such a study.

1-8. [5] Traditional polling operations like Gallup use a procedure called random digit dialing, which dials random strings of digits instead of picking phone numbers from the phone book. Suggest why such polls are conducted using random digit dialing.

Implementation Projects 1-9. [5] Write a program to scrape the best-seller rank for a book on Amazon.com. Use this to plot the rank of all of Skiena's books over time. Which one of these books should be the next item that you purchase? Do you have friends for whom they would make a welcome and appropriate gift? :-)

1-10. [5] For your favorite sport (baseball, football, basketball, cricket, or soccer) identify a data set with the historical statistical records for all major participants. Devise and implement a ranking system to identify the best player at each position.

Interview Questions 1-11. [3] For each of the following questions: (1) produce a quick guess based only on your understanding of the world, and then (2) use Google to find supportable numbers to produce a more principled estimate from. How much did your two estimates differ by?
(a) How many piano tuners are there in the entire world?
(b) How much does the ice in a hockey rink weigh?
(c) How many gas stations are there in the United States?

(d) How many people fly in and out of LaGuardia Airport every day?
(e) How many gallons of ice cream are sold in the United States each year?
(f) How many basketballs are purchased by the National Basketball Association (NBA) each year?
(g) How many fish are there in all the world's oceans?
(h) How many people are flying in the air right now, all over the world?
(i) How many ping-pong balls can fit in a large commercial jet?
(j) How many miles of paved road are there in your favorite country?

(k) How many dollar bills are sitting in the wallets of all people at Stony Brook University?
(l) How many gallons of gasoline does a typical gas station sell per day?
(m) How many words are there in this book?
(n) How many cats live in New York city?
(o) How much would it cost to fill a typical car's gas tank with Starbuck's coffee?
(p) How much tea is there in China?
(q) How many checking accounts are there in the United States?

1-12. [3] What is the difference between regression and classification?
1-13. [8] How would you build a data-driven recommendation system? What are the limitations of this approach?
1-14. [3] How did you become interested in data science?
1-15. [3] Do you think data science is an art or a science?

Kaggle Challenges 1-16. Who survived the shipwreck of the Titanic?
https://www.kaggle.com/c/titanic
$1-17$. Where is a particular taxi cab going?
https://www.kaggle.com/c/pkdd-15-predict-taxi-service-trajectory-i
1-18. How long will a given taxi trip take?
https://www.kaggle.com/c/pkdd-15-taxi-trip-time-prediction-ii

Chapter 2

# Chapter 4

# Mathematical Preliminaries

A data scientist is someone who *knows* more *statistics* than a computer scientist and more computer science than a statistician.

- Josh Blumenstock

You must walk before you can run. Similarly, there is a certain level of mathematical maturity which is necessary before you should be trusted to do anything meaningful with numerical data.

In writing this book, I have assumed that the reader has had some degree of exposure to *probability and statistics*, linear algebra, and continuous mathematics. I have also assumed that they have probably forgotten most of it, or perhaps didn't *always* see the forest (why things are important, and how to use them) for the trees (all the details of definitions, proofs, and operations).

This chapter will try to refresh your understanding of certain basic mathematical concepts. Follow along with me, and pull out your old textbooks if necessary for future reference. Deeper concepts will be introduced later in the book when we need them.

## 4.1 Probability

Probability theory provides a formal framework for reasoning about the likelihood of events. Because it is a formal discipline, there are a thicket of associated definitions to instantiate exactly what we are reasoning about:

- An experiment is a procedure which yields one of a set of possible outcomes. As our ongoing example, consider the experiment of tossing two six-sided dice, one red and one blue, with each face baring a distinct integer $\{1, \ldots, 6\}$.

- A sample space $S$ is the set of possible outcomes of an experiment. In our dice example, there are 36 possible outcomes, namely

$$\begin{aligned}S = \{&(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),\\&(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\\&(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}.\end{aligned}$$

- An *event* $E$ *is a specified subset of the outcomes of an experiment. The event that the sum of the dice equals 7 or 11 (the conditions to win at craps on the first roll) is the subset*

$$E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1),(5,6),(6,5)\}$$

- The probability of an outcome $s$, denoted $p(s)$ is a number with the two properties:
- For each outcome $s$ in sample space $S, 0 \leq p(s) \leq 1$.
- The sum of probabilities of all outcomes adds to one: $\sum_{s \in S} p(s) = 1$.

If we assume two distinct fair dice, the probability $p(s) = (1/6) \times (1/6) = 1/36$ for all outcomes $s \in S$.

- The probability of an event $E$ is the sum of the probabilities of the outcomes of the experiment. Thus

$$p(E) = \sum_{s \in E} p(s)$$

An alternate formulation is in terms of the *complement* of the event $\bar{E}$, the case when $E$ does not occur. Then

$$P(E) = 1 - P(\bar{E}).$$

This is useful, because often it is easier to analyze $P(\bar{E})$ than $P(E)$ directly.

- A *random variable* $V$ *is a numerical function on the outcomes of a probability space. The function "sum the values of two dice"* $(V((a,b)) = a + b)$ *produces an integer result between 2 and 12. This implies a probability distribution of the values of the random variable. The probability* $P(V(s) = 7) = 1/6$*, as previously shown, while* $P(V(s) = 12) = 1/36$.

- *The expected value of a random variable* $V$ *defined on a sample space* $S$*,* $E(V)$ *is defined*

$$E(V) = \sum_{s \in S} p(s) \cdot V(s)$$

All this you have presumably seen before. But it provides the language we will use to connect between probability and statistics. The data we see usually comes from measuring properties of observed events. The theory of probability and statistics provides the tools to analyze this data.

Probability vs. *Statistics* Probability and statistics are related areas of mathematics which concern themselves with analyzing the relative frequency of events. Still, there are fundamental differences in the way they see the world:

Probability deals with predicting the likelihood of future events, while statistics involves the analysis of the frequency of past events.

- Probability is primarily a theoretical branch of mathematics, which studies the consequences of mathematical definitions. Statistics is primarily an applied branch of mathematics, which tries to make sense of observations in the real world.

Both subjects are important, relevant, and useful. But they are different, and understanding the distinction is crucial in properly interpreting the relevance of mathematical evidence. Many a gambler has gone to a cold and lonely grave for failing to make the proper distinction between probability and statistics.

This distinction will perhaps become clearer if we trace the thought process of a mathematician encountering her first craps game:

- If this mathematician were a probabilist, she would see the dice and think "Six-sided dice? Each side of the dice is presumably equally likely to land face up. Now *assuming* that each face comes up with probability $1/6$, I can figure out what my chances are of crapping out."

- If instead a statistician wandered by, she would see the dice and think "How do I know that they are not loaded? I'll watch a while, and keep track of how often each number comes up. Then I can decide if my observations are consistent with the assumption of equal-probability faces. Once I'm confident enough that the dice are fair, I'll call a probabilist to tell me how to bet."

In summary, probability theory enables us to find the consequences of a given ideal world, while statistical theory enables us to measure the extent to which our world is ideal. This constant tension between theory and practice is why statisticians prove to be a tortured group of individuals compared with the happy-go-lucky probabilists.

Modern probability theory first emerged from the dice tables of France in 1654. Chevalier de Méré, a French nobleman, wondered whether the player or the house had the advantage in a particular betting game. [1] In the basic version, the player rolls four dice, and wins provided none of them are a 6 . The house collects on the even money bet if at least one 6 appears.

De Méré brought this problem to the attention of the French mathematicians Blaise Pascal and Pierre de Fermat, most famous as the source of Fermat's Last Theorem. Together, these men worked out the basics of probability theory, along the way establishing that the house wins this dice game with proba-

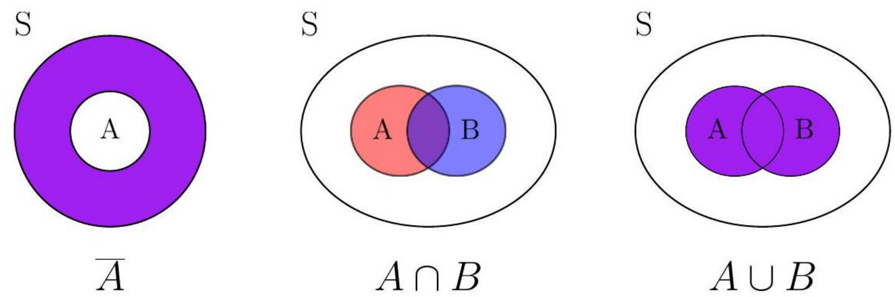


Figure 4.1: Venn diagrams illustrating *set difference* (left), *intersection* (middle), and *union* (right).

bility $p = 1 - (5/6)^4 \approx 0.517$, where the probability $p = 0.5$ would denote a fair game where the house wins exactly half the time.

### 4.1.1 Compound Events and Independence

We will be interested in complex events computed from simpler events $A$ and $B$ on the same set of outcomes. Perhaps event $A$ is that at least one of two dice be an even number, while event $B$ denotes rolling a total of either 7 or 11. Note that there exist certain outcomes of $A$ which are not outcomes of $B$, specifically

$$A - B = \{(1,2),(1,4),(2,1),(2,2),(2,3),(2,4),(2,6),(3,2),(3,6),(4,1)$$
$$(4,2),(4,4),(4,5),(4,6),(5,4),(6,2),(6,3),(6,4),(6,6)\}$$

This is the set difference operation. Observe that here $B - A = \{\}$, because every pair adding to 7 or 11 must contain one odd and one even number.

The outcomes in common between both events $A$ and $B$ are called the intersection, denoted $A \cap B$. This can be written as

$$A \cap B = A - (S - B).$$

Outcomes which appear in either $A$ or $B$ are called the union, denoted $A \cup B$. With the complement operation $\bar{A} = S - A$, we get a rich language for combining events, shown in Figure 2.1. We can readily compute the probability of any of these sets by summing the probabilities of the outcomes in the defined sets.

[1] He really shouldn't have wondered. The house always has the advantage.

The events $A$ and $B$ are *independent* if and only if

$$P(A \cap B) = P(A) \times P(B)$$

This means that there is no special structure of outcomes shared between events $A$ and $B$. Assuming that half of the students in my class are female, and half the students in my class are above average, we would expect that a quarter of my students are both female and above average if the events are independent.

Probability theorists love independent events, because it simplifies their calculations. But data scientists generally don't. When building models to predict the likelihood of some future event $B$, given knowledge of some previous event $A$, we want as strong a dependence of $B$ on $A$ as possible.

Suppose I always use an umbrella if and only if it is raining. Assume that the probability it is raining here (event $B$ ) is, say, $p = 1/5$. This implies the probability that I am carrying my umbrella (event $A$ ) is $q = 1/5$. But even more, if you know the state of the rain you know exactly whether I have my umbrella. These two events are perfectly *correlated*.

By contrast, suppose the events were independent. Then

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

and whether it is raining has absolutely no impact on whether I carry my protective gear.

Correlations are the driving force behind predictive models, so we will discuss how to measure them and what they mean in Section 2.3

### 4.1.2 Conditional Probability

When two events are correlated, there is a dependency between them which makes calculations more difficult. The *conditional probability* of $A$ given $B$, $P(A \mid B)$ is defined:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Recall the dice rolling events from Section 2.1.2 namely:

- Event $A$ is that at least one of two dice be an even number.
- Event $B$ is the sum of the two dice is either a 7 or an 11 .

Observe that $P(A \mid B) = 1$, because any roll summing to an odd value must consist of one even and one odd number. Thus $A \cap B = B$, analogous to the umbrella case above. For $P(B \mid A)$, note that $P(A \cap B) = 9/36$ and $P(A) = 25/36$, so $P(B \mid A) = 9/25$.

Conditional probability will be important to us, because we are interested in the likelihood of an event $A$ (perhaps that a particular piece of email is spam) as a function of some evidence $B$ (perhaps the distribution of words within the document). Classification problems generally reduce to computing conditional probabilities, in one way or another.

Our primary tool to compute conditional probabilities will be *Bayes theorem*, which reverses the direction of the dependencies:

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}$$

Often it proves easier to compute probabilities in one direction than another, as in this problem. By Bayes theorem $P(B \mid A) = (1 \cdot 9/36)/(25/36) = 9/25$, exactly

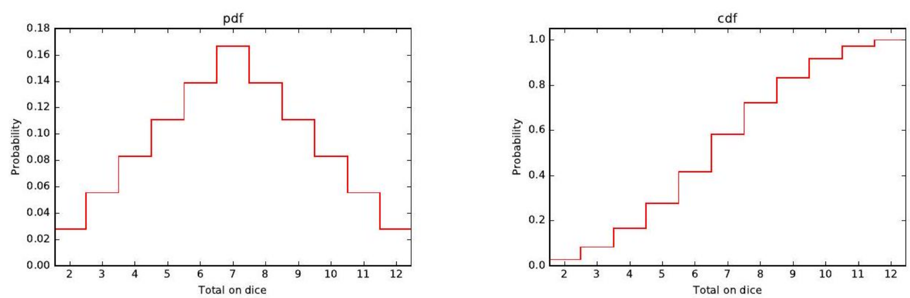


Figure 4.2: The *probability density function* (pdf) of the sum of two dice contains *exactly* the same information as the *cumulative density function* (cdf), but looks very different.

what we got before. We will revisit Bayes theorem in Section 5.6. where it will establish the foundations of computing probabilities in the face of evidence.

### 4.1.3 Probability Distributions

Random variables are numerical functions where the values are associated with probabilities of occurrence. In our example where $V(s)$ the sum of two tossed dice, the function produces an integer between 2 and 12. The probability of a particular value $V(s) = X$ is the sum of the probabilities of all the outcomes which add up to $X$.

Such random variables can be represented by their probability density function, or pdf. This is a graph where the $x$-axis represents the range of values the random variable can take on, and the $y$-axis denotes the probability of that given value. Figure 2.2 (left) presents the pdf of the sum of two fair dice. Observe that the peak at $X = 7$ corresponds to the most frequent dice total, with a probability of $1/6$.

Such pdf plots have a strong relationship to histograms of data frequency, where the $x$-axis again represents the range of value, but $y$ now represents the observed frequency of exactly how many event occurrences were seen for each given value $X$. Converting a histogram to a pdf can be done by dividing each bucket by the total frequency over all buckets. The sum of the entries then becomes 1 , so we get a probability distribution.

Histograms are statistical: they reflect actual observations of outcomes. In contrast, pdfs are probabilistic: they represent the underlying chance that the next observation will have value $X$. We often use the histogram of observations $h(x)$ in practice to estimate the probabilities [2] by normalizing counts by the total

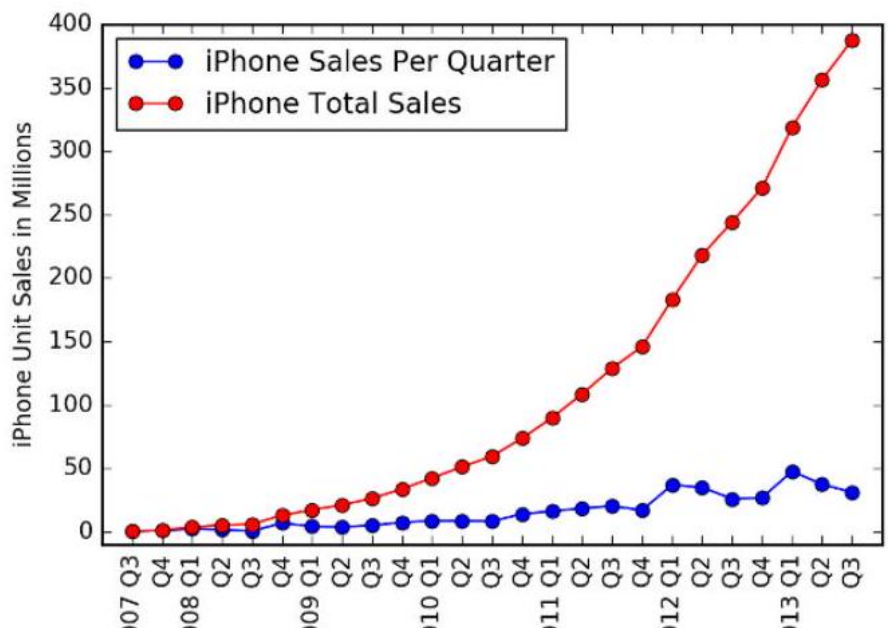


Figure 4.3: iPhone quarterly sales data presented as cumulative and incremental (quarterly) distributions. Which curve did Apple CEO Tim Cook choose to present?

number of observations:

$$P(k = X) = \frac{h(k = X)}{\sum_x h(x = X)}$$

There is another way to represent random variables which often proves useful, called a cumulative density function or cdf. The cdf is the running sum of

[2] A technique called *discounting* offers a better way to estimate the frequency of rare events, and will be discussed in Section 11.1 .2

the probabilities in the pdf; as a function of $k$, it reflects the probability that $X \leq k$ instead of the probability that $X = k$. Figure 2.2 (right) shows the cdf of the dice sum distribution. The values increase monotonically from left to right, because each term comes from adding a positive probability to the previous total. The rightmost value is 1 , because all outcomes produce a value no greater than the maximum.

It is important to realize that the pdf $P(V)$ and $\mathrm{cdf} C(V)$ of a given random variable $V$ contain exactly the same information. We can move back and forth between them because:

$$P(k = X) = C(X \leq k + \delta) - C(X \leq k)$$

where $\delta = 1$ for integer distributions. The cdf is the running sum of the pdf, so

$$C(X \leq k) = \sum_{x \leq k} P(X = x)$$

Just be aware of which distribution you are looking at. Cumulative distributions always get higher as we move to the right, culminating with a probability of $C(X \leq \infty) = 1$. By contrast, the total area under the curve of a pdf equals 1 , so the probability at any point in the distribution is generally substantially less.

An amusing example of the difference between cumulative and incremental distributions is shown in Figure 2.3 Both distributions show *exactly* the same data on Apple iPhone sales, but which curve did Apple CEO Tim Cook choose to present at a major shareholder event? The cumulative distribution (red) shows that sales are exploding, right? But it presents a misleading view of growth rate, because incremental change is the derivative of this function, and hard to visualize. Indeed, the sales-per-quarter plot (blue) shows that the rate of iPhone sales actually had declined for the last two periods before the presentation.

## 4.2 Descriptive Statistics

Descriptive statistics provide ways of capturing the properties of a given data set or sample. They summarize observed data, and provide a language to talk about it. Representing a group of elements by a new derived element, like *mean*, min, count, or sum reduces a large data set to a small summary statistic: aggregation as data reduction.

Such statistics can become features in their own right when taken over natural groups or clusters in the full data set. There are two main types of descriptive statistics:

- *Central tendency measures*, which capture the center around which the data is distributed.
- *Variation* or *variability measures*, which describe the data spread, i.e. how far the measurements lie from the center.

Together these statistics tell us an enormous amount about our distribution.

### 4.2.1 Centrality Measures

The first element of statistics we are exposed to in school are the basic centrality measures: mean, *median*, and mode. These are the right place to start when thinking of a single number to characterize a data set.

- Mean: You are probably quite comfortable with the use of the arithmetic mean, where we sum values and divide by the number of observations:

$$\mu_X = \frac{1}{n}\sum_{i=1}^{n} x_i$$

We can easily maintain the mean under a stream of insertions and deletions, by keeping the sum of values separate from the frequency count, and divide only on demand.
The mean is very meaningful to characterize symmetric distributions without outliers, like height and weight. That it is symmetric means the number of items above the mean should be roughly the same as the number

below. That it is without outliers means that the range of values is reasonably tight. Note that a single MAXINT creeping into an otherwise sound set of observations throws the mean wildly off. The median is a centrality measure which proves more appropriate with such ill-behaved distributions.

- *Geometric mean*: The *geometric mean* is the $n$th root of the product of $n$ values:

$$\left(\prod_{i=1}^{n} a_i\right)^{1/n} = \sqrt[n]{a_1 a_2 \dots a_n}$$

The geometric mean is always less than or equal to the arithmetic mean. For example, the geometric mean of the sums of 36 dice rolls is 6.5201 , as opposed to the arithmetic mean of 7 . It is very sensitive to values near zero. A single value of zero lays waste to the geometric mean: no matter what other values you have in your data, you end up with zero. This is somewhat analogous to having an outlier of $\infty$ in an arithmetic mean.
But geometric means prove their worth when averaging ratios. The geometric mean of $1/2$ and $2/1$ is 1 , whereas the mean is 1.25 . There is less available "room" for ratios to be less than 1 than there is for ratios above 1 , creating an asymmetry that the arithmetic mean overstates. The geometric mean is more meaningful in these cases, as is the arithmetic mean of the *logarithms* of the ratios.

- Median: The median is the exact middle value among a data set; just as many elements lie above the median as below it. There is a quibble about what to take as the median when you have an even number of elements. You can take either one of the two central candidates: in any reasonable data set these two values should be about the same. Indeed in the dice example, both are 7 .
  A nice property of the median as so defined is that it must be a genuine value of the original data stream. There actually is someone of median height to you can point to as an example, but presumably no one in the world is of exactly average height. You lose this property when you average the two center elements.
  Which centrality measure is best for applications? The median typically lies pretty close to the arithmetic mean in symmetrical distributions, but it is often interesting to see how far apart they are, and on which side of the mean the median lies.
  The median generally proves to be a better statistic for skewed distributions or data with outliers: like wealth and income. Bill Gates adds $250 to the mean per capita wealth in the United States, but nothing to the median. If he makes you personally feel richer, then go ahead and use the mean. But the median is the more informative statistic here, as it will be for any power law distribution.

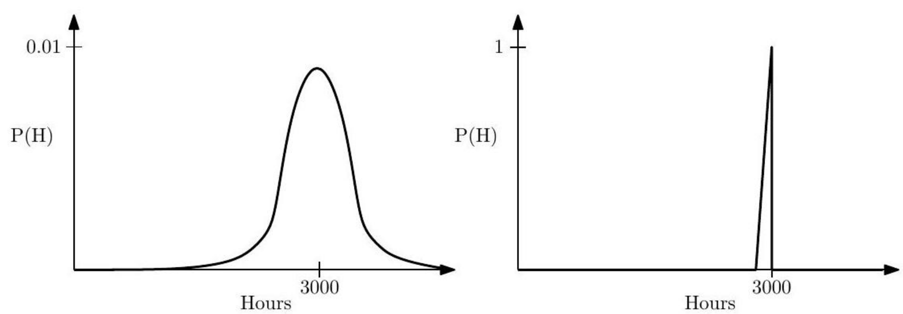


Figure 4.4: normal (left) and with zero *variance (right*).

- *Mode*: The *mode* is the most frequent element in the data set. This is 7 in our ongoing dice example, because it occurs six times out of thirty-six elements. Frankly, I've never seen the mode as providing much insight as centrality measure, because it often isn't close to the center. Samples measured over a large range should have very few repeated elements or collisions at any particular value. This makes the mode a matter of happenstance. Indeed, the most frequently occurring elements often reveal artifacts or anomalies in a data set, such as default values or error codes that do not really represent elements of the underlying distribution.
  The related concept of the peak in a frequency distribution (or histogram)

is meaningful, but interesting peaks only get revealed through proper bucketing. The current peak of the annual salary distribution in the United States lies between $\$30,000$ and $\$40,000$ per year, although the mode presumably sits at zero.

### 4.2.2 Variability Measures

The most common measure of variability is the standard deviation $\sigma$, which measures sum of squares differences between the individual elements and the mean:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(a_i - \bar{a})^2}{n-1}}$$

A related statistic, the variance $V$, is the square of the standard deviation, i.e. $V = \sigma^2$. Sometimes it is more convenient to talk about variance than standard deviation, because the term is eight characters shorter. But they measure exactly the same thing.

As an example, consider the humble light bulb, which typically comes with an expected working life, say $\mu = 3000$ hours, derived from some underlying distribution shown in Figure 2.4 In a conventional bulb, the chance of it lasting longer than $\mu$ is presumably about the same as that of it burning out quicker, and this degree of uncertainty is measured by $\sigma$. Alternately, imagine a "printer

cartridge bulb," where the evil manufacturer builds very robust bulbs, but includes a counter so they can prevent it from ever glowing after 3000 hours of use. Here $\mu = 3000$ and $\sigma = 0$. Both distributions have the same mean, but substantially different variance.

The sum of squares penalty in the formula for $\sigma$ means that one outlier value $d$ units from the mean contributes as much to the variance as $d^2$ points each one unit from the mean, so the variance is very sensitive to outliers.

An often confusing matter concerns the denominator in the formula for standard deviation. Should we divide by $n$ or $n-1$ ? The difference here is technical. The standard deviation of the full *population* divides by $n$, whereas the standard deviation of the *sample* divides by $n-1$. The issue is that sampling just one point tells us absolutely nothing about the underlying variance in any population, where it is perfectly reasonable to say there is zero variance in weight among the population of a one-person island. But for reasonable-sized data sets $n \approx (n-1)$, so it *really* doesn't matter.

### 4.2.3 Interpreting Variance

Repeated observations of the same phenomenon do not always produce the same results, due to random noise or error. *Sampling errors* result when our observations capture unrepresentative circumstances, like measuring rush hour traffic on weekends as well as during the work week. *Measurement errors* reflect the limits of precision inherent in any sensing device. The notion of signal to noise ratio captures the degree to which a series of observations reflects a quantity of interest as opposed to data variance. As data scientists, we care about changes in the signal instead of the noise, and such variance often makes this problem surprisingly difficult.

I think of variance as an inherent property of the universe, akin to the speed of light or the time-value of money. Each morning you weigh yourself on a scale you are guaranteed to get a different number, with changes reflecting when you last ate (sampling error), the flatness of the floor, or the age of the scale (both measurement error) as much as changes in your body mass (actual variation). So what is your real weight?

Every measured quantity is subject to some level of variance, but the phenomenon cuts much deeper than that. Much of what happens in the world is just random fluctuations or arbitrary happenstance causing variance even when the situation is unchanged. Data scientists seek to explain the world through data, but distressingly often there is no real phenomena to explain, only a ghost created by variance. Examples include:

- *The stock market*: Consider the problem of measuring the relative "skill" of different stock market investors. We know that Warren Buffet is much better at investing than we are. But very few professional investors prove consistently better than others. Certain investment vehicles wildly outperform the market in any given time period. However, the hot fund one

```
ln[28]:= Season[p_Real, n_Integer] :=
  Count[Table[If[RandomReal[1] }\leqslantP,1,0],{n}], 1]/(1.0*n
In[29]:= Histogram[d = Table[Season[0.300, 500], {100 000}], 100]
```

year usually underperforms the market the year after, which shouldn't happen if this outstanding performance was due to skill rather than luck.

The fund managers themselves are quick to credit profitable years to their own genius, but losses to unforeseeable circumstances. However, several studies have shown that the performance of professional investors is essentially random, meaning there is little real difference in skill. Most investors are paying managers for previously-used luck. So why do these entrail-readers get paid so much money?

- *Sports performance*: Students have good semesters and bad semesters, as reflected by their grade point average (GPA). Athletes have good and bad seasons, as reflected by their performance and statistics. Do such changes reflect genuine differences in effort and ability, or are they just variance?

  In baseball, .300 hitters (players who hit with a $30\%$ success rate) represent consistency over a full season. Batting .275 is not a noteworthy

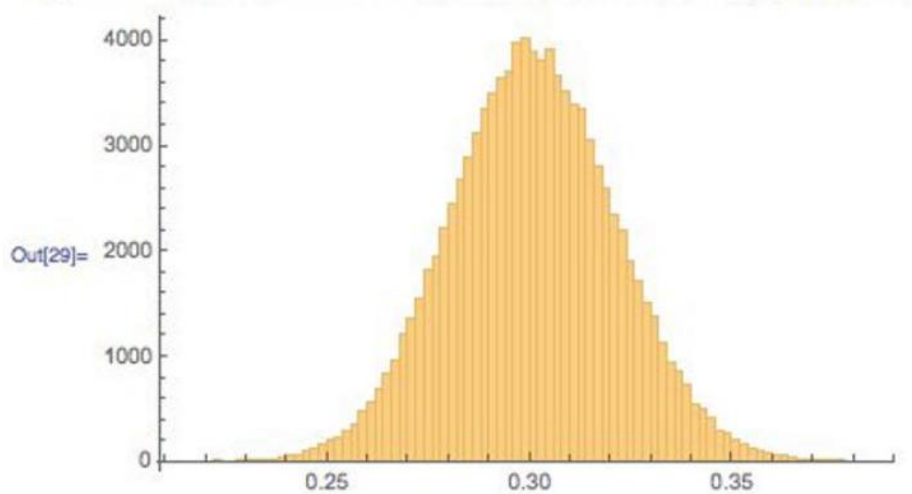


Figure 4.5: Sample variance on hitters with a real 30% success rate results in a wide range of observed performance even over 500 trials per season.

season, but hit .300 and you are a star. Hit . 325 and you are likely to be the batting champion.

Figure 2.5 shows the results of a simple simulation, where random numbers were used to decide the outcome of each at-bat over a 500 at-bats/season. Our synthetic player is a real .300 hitter, because we programmed it to report a hit with probability 300/1000 (0.3). The results show that a real . 300 hitter has a 10% chance of hitting .275 or below, just by chance. Such a season will typically be explained away by injuries or maybe the inevitable effects of age on athletic performance. But it could just be natural variance. Smart teams try to acquire a good hitter after a lousy season, when the price is cheaper, trying to take advantage of this variance.

Our . 300 hitter also has a 10% chance of batting above .325, but you

can be pretty sure that they will ascribe such a breakout season to their improved conditioning or training methods instead of the fact they just got lucky. Good or bad season, or lucky/unlucky: it is hard to tell the signal from the noise.

- *Model performance*: As data scientists, we will typically develop and evaluate several models for each predictive challenge. The models may range from very simple to complex, and vary in their training conditions or parameters.

Typically the model with the best accuracy on the training corpus will be paraded triumphantly before the world as the right one. But small differences in

the performance between models is likely explained by simple variance rather than wisdom: which training/evaluation pairs were selected, how well parameters were optimized, etc.
Remember this when it comes to training machine learning models. Indeed, when asked to choose between models with small performance differences between them, I am more likely to argue for the simplest model than the one with the highest score. Given a hundred people trying to predict heads and tails on a stream of coin tosses, one of them is guaranteed to end up with the most right answers. But there is no reason to believe that this fellow has any better predictive powers than the rest of us.

### 4.2.4 Characterizing Distributions

Distributions do not necessarily have much probability mass exactly at the mean. Consider what your wealth would look like after you borrow $\$100$ million, and then bet it all on an even money coin flip. Heads you are now $\$100$ million in clear, tails you are $\$100$ million in hock. Your expected wealth is zero, but this mean does not tell you much about the shape of your wealth distribution.

However, taken together the mean and standard deviation do a decent job of characterizing any distribution. Even a relatively small amount of mass positioned far from the mean would add a lot to the standard deviation, so a small value of $\sigma$ implies the bulk of the mass must be near the mean.

To be precise, regardless of how your data is distributed, at least ( $1-\ \left(1/k^2\right)$) th of the mass must lie within $\pm k$ standard deviations of the mean. This means that at least $75\%$ of all the data must lie within $2\sigma$ of the mean, and almost $89\%$ within $3\sigma$ for any distribution.

We will see that even tighter bounds hold when we know the distribution is well-behaved, like the Gaussian or normal distribution. But this is why it is a great practice to report both $\mu$ and $\sigma$ whenever you talk about averages. The average height of adult women in the United States is $63.7 \pm 2.7$ inches, meaning $\mu = 63.7$ and $\sigma = 2.7$. The average temperature in Orlando, Fl is 60.3 degrees Fahrenheit. However, there have been many more 100 degree days at Disney World than 100 inch ( 8.33 foot) women visiting to enjoy them.

*Take-Home Lesson*: Report both the mean and standard deviation to characterize your distribution, written as $\mu \pm \sigma$.

## 4.3 Correlation Analysis

Suppose we are given two variables $x$ and $y$, represented by a sample of $n$ points of the form $(x_i, y_i)$, for $1 \leq i \leq n$. We say that $x$ and $y$ are *correlated* when the value of $x$ has some predictive power on the value of $y$.

The correlation coefficient $r(X, Y)$ is a statistic that measures the degree to which $Y$ is a function of $X$, and vice versa. The value of the *correlation coefficient ranges* from -1 to 1 , where 1 means fully correlated and 0 implies no relation, or independent variables. Negative correlations imply that the variables are *anti-correlated*, meaning that when $X$ goes up, $Y$ goes down.

Perfectly anti-correlated variables have a correlation of -1 . Note that negative correlations are just as good for predictive purposes as positive ones. That you are less likely to be unemployed the more education you have is an example of a negative correlation, so the level of education can indeed help predict job status. Correlations around 0 are useless for forecasting.

Observed correlations drives many of the predictive models we build in data science. Representative strengths of correlations include:

- Are taller people more likely to remain lean? The observed correlation between height and BMI is $r = -0.711$, so height is indeed negatively correlated with body mass index (BMI) [3]

- Do standardized tests predict the performance of students in college? The observed correlation between SAT scores and freshmen GPA is $r = 0.47$, so yes, there is some degree of predictive power. But social economic status is just as strongly correlated with SAT scores $(r = 0.42)$ ⌴[4]

- Does financial status affect health? The observed correlation between household income and the prevalence of coronary artery disease is $r =$ -0.717 , so there is a strong negative correlation. So yes, the wealthier you are, the lower your risk of having a heart attack. $5$[5]

- Does smoking affect health? The observed correlation between a group's propensity to smoke and their mortality rate is $r = 0.716$, so for G-d's sake, don't smoke [6]

- Do violent video games increase aggressive behavior? The observed correlation between play and violence is $r = 0.19$, so there is a weak but significant correlation. 7

This section will introduce the primary measurements of correlation. Further, we study how to appropriately determine the strength and power of any observed correlation, to help us understand when the connections between variables are real.

Correlation Coefficients: *Pearson and Spearman Rank* In fact, there are two primary statistics used to measure correlation. Mercifully, both operate on the same -1 to 1 scale, although they measure somewhat different things. These different statistics are appropriate in different situations, so you should be aware of both of them.

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$\sqrt[3]{}$ https://onlinecourses.science.psu.edu/stat500/node/60

4 https://research.collegeboard.org/sites/default/files/publications/2012/9/ researchreport-2009-1-socioeconomic-status-sat-freshman-gpa-analysis-data.pdf

° http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3457990/

6 http://lib.stat.cmu.edu/DASL/Stories/SmokingandCancer.html

**The Pearson Correlation Coefficient** The more prominent of the two statistics is Pearson correlation, defined as

$$r = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}\sqrt{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2}} = \frac{\mathrm{Cov}(X,Y)}{\sigma(X)\sigma(Y)}$$

Let's parse this equation. Suppose $X$ and $Y$ are strongly correlated. Then we would expect that when $x_i$ is greater than the mean $\bar{X}$, then $y_i$ should be bigger than its mean $\bar{Y}$. When $x_i$ is lower than its mean, $y_i$ should follow. Now look at the numerator. The sign of each term is positive when both values are above $(1 \times 1)$ or below $(-1 \times -1)$ their respective means. The sign of each term is negative $((-1 \times 1)$ or $(1 \times -1))$ if they move in opposite directions, suggesting negative correlation. If $X$ and $Y$ were uncorrelated, then positive and negative terms should occur with equal frequency, offsetting each other and driving the value to zero.

The numerator's operation determining the sign of the correlation is so useful that we give it a name, *covariance*, computed:

$$\mathrm{Cov}(X,Y) = \sum_{i=1}^{n}\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)$$

Remember covariance: we will see it again in Section 8.2.3
The denominator of the Pearson formula reflects the amount of variance in the two variables, as measured by their standard deviations. The covariance between $X$ and $Y$ potentially increases with the variance of these variables, and this denominator is the magic amount to divide it by to bring correlation to a -1 to 1 scale.

Anderson-Aggression.pdf

**The Spearman Rank Correlation Coefficient** The Pearson correlation coefficient defines the degree to which a linear predictor of the form $y = m \cdot x + b$ can fit the observed data. This generally does a good job measuring the similarity between the variables, but it is possible to construct pathological examples where the correlation coefficient between $X$ and $Y$ is zero, yet $Y$ is completely dependent on (and hence perfectly predictable from) $X$.

Consider points of the form $(x, |x|)$, where $x$ is uniformly (or symmetrically) sampled from the interval $[-1, 1]$ as shown in Figure 2.6 The correlation will be zero because for every point $(x, x)$ there will be an offsetting point $(-x, x)$, yet $y = |x|$ is a perfect predictor. Pearson correlation measures how well the best linear predictors can work, but says nothing about weirder functions like absolute value.

The *Spearman rank correlation coefficient* essentially counts the number of pairs of input points which are out of order. Suppose that our data set contains points $(x_1, y_1)$ and $(x_2, y_2)$ where $x_1 < x_2$ and $y_1 < y_2$. This is a vote that

[7] http://webspace.pugetsound.edu/facultypages/cjones/chidev/Paper/Articles/

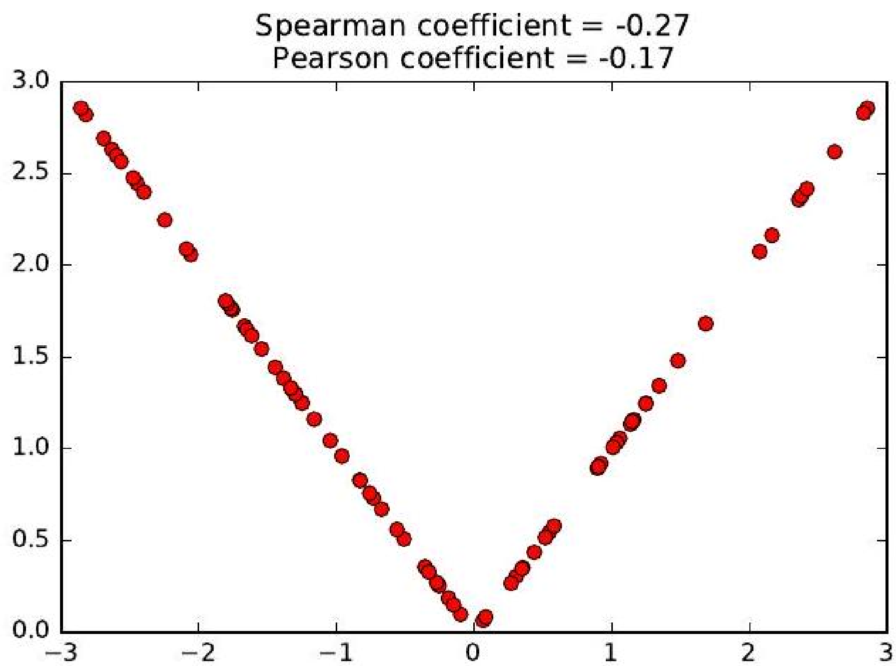



Figure 4.6: The function $y = |x|$ does not have a *linear* model, but seems like it should be easily fitted despite weak correlations.

the values are positively correlated, whereas the vote would be for a negative correlation if $y_2 < y_1$.

Summing up over all pairs of points and normalizing properly gives us Spearman rank correlation. Let $\operatorname{rank}(x_i)$ be the rank position of $x_i$ in sorted order among all $x_i$, so the rank of the smallest value is 1 and the largest value $n$. Then

$$\rho = 1 - \frac{6 \sum d_i^2}{n\left(n^2-1\right)}$$

where $d_i = \operatorname{rank}(x_i) - \operatorname{rank}(y_i)$.

The relationship between our two coefficients is better delineated by the example in Figure 2.7. In addition to giving high scores to non-linear but monotonic functions, Spearman correlation is less sensitive to extreme outlier elements than Pearson. Let $p = (x_1, y_{\max})$ be the data point with largest value

of $y$ in a given data set. Suppose we replace $p$ with $p' = (x_1, \infty)$. The Pearson correlation will go crazy, since the best fit now becomes the vertical line $x = x_1$. But the Spearman correlation will be unchanged, since all the points were under $p$, just as they are now under $p'$.

The Power and Significance of Correlation The correlation coefficient $r$ reflects the degree to which $x$ can be used to predict $y$ in a given sample of points $S$. As $|r| \rightarrow 1$, these predictions get better and better.

But the real question is how this correlation will hold up in the real world, outside the sample. Stronger correlations have larger $|r|$, but also involve samples of enough points to be *significant*. There is a wry saying that if you want to fit your data by a straight line, it is best to sample it at only two points. Your correlation becomes more impressive the more points it is based on.

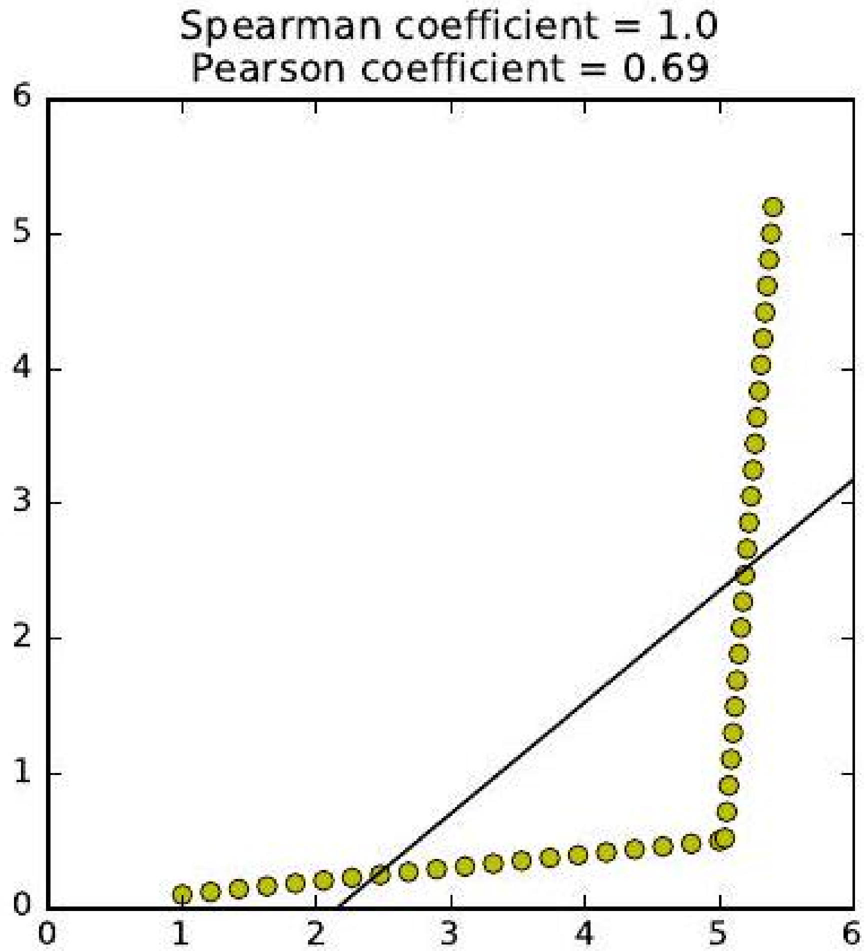
The statistical limits in interpreting correlations are presented in Figure 2.8 based on strength and size:

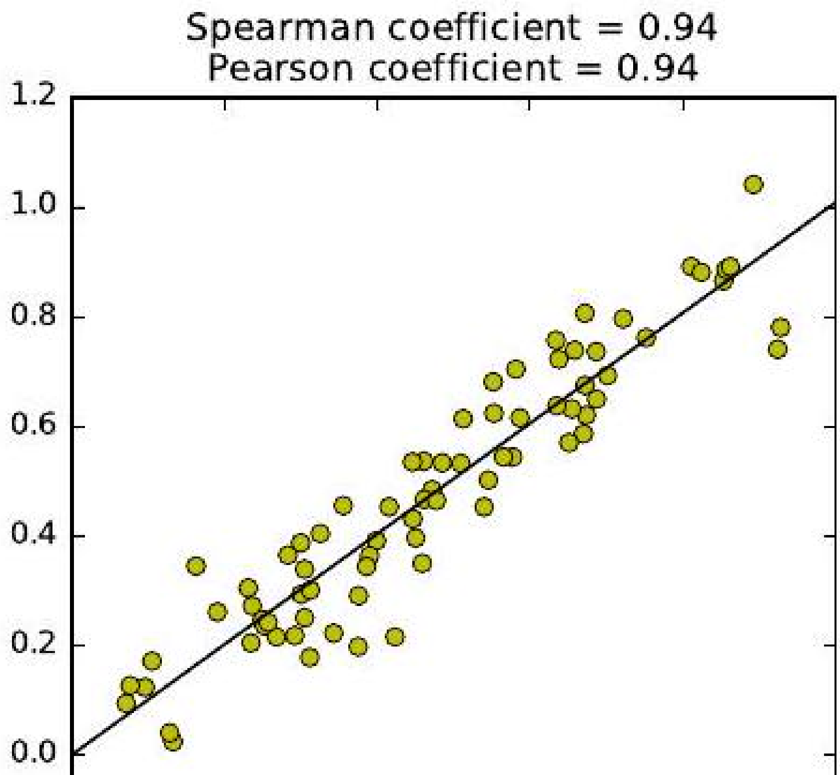
- Strength of correlation: $R^2$ : The square of the sample correlation coefficient $r^2$ estimates the fraction of the variance in $Y$ explained by $X$ in a simple linear regression. The correlation between height and weight is approximately 0.8 , meaning it explains about two thirds of the variance.

What do we mean by "explaining the variance"? Let $f(x) = mx + c$ be the

Figure 2.9: Plotting $r_i = y_i - f(x_i)$ shows that the *residual* values have lower variance and mean zero. The original data points are on the left, with the corresponding residuals on the right.

predictive value of $y$ from $x$, with the parameters $m$ and $c$ corresponding to the best possible fit. The residual values $r_i = y_i - f(x_i)$ will have mean zero, as shown in Figure 2.9 . Further, the variance of the full data set $V(Y)$ should be much larger than $V(r)$ if there is a good linear fit $f(x)$. If $x$ and $y$ are perfectly correlated, there should be no residual error, and $V(r) = 0$. If $x$ and $y$ are totally uncorrelated, the fit should contribute nothing, and $V(y) \approx V(r)$. Generally speaking, $1 - r^2 = V(r)/V(y)$.


Spearman coefficient = 1.0
Pearson coefficient = 0.69
0
1
2
3
4
5
6


Spearman coefficient = 0.94
Pearson coefficient = 0.94
0.0
0.2
0.4
0.6
0.8
1.0
1.2

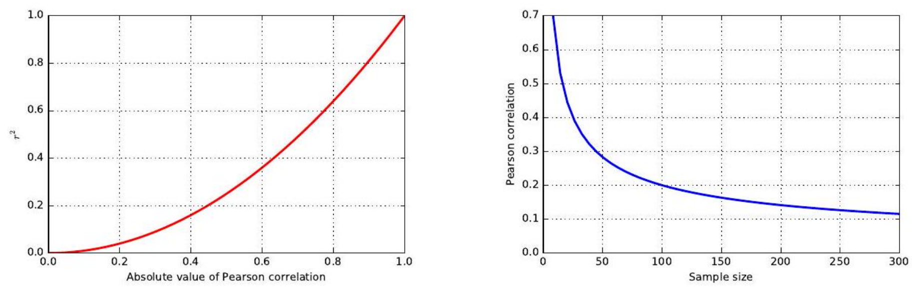



Figure 4.8: (left) shows how rapidly $r^2$ decreases with $r$. There is a profound limit to how excited we should get about establishing a weak correlation. A correlation of 0.5 possesses only $25\%$ of the maximum predictive power, and a correlation of $r = 0.1$ only $1\%$. Thus the predictive value of correlations decreases rapidly with $r$.

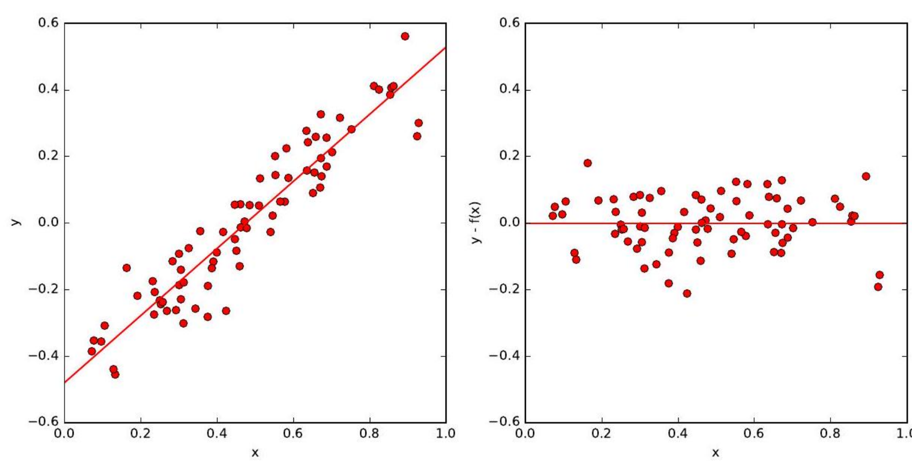



Figure 4.9: Limits in interpreting significance. The $r^2$ value shows that weak correlations explain only a small fraction of the variance (left). The level of correlation necessary to be statistically significance decreases rapidly with sample size $n$ (right).

Consider Figure 2.9, showing a set of points (left) admitting a good linear fit, with correlation $r = 0.94$. The corresponding residuals $r_i = y_i - f(x_i)$ are plotted on the right. The variance of the $y$ values on the left $V(y) =$ 0.056 , substantially greater than the variance $V(r) = 0.0065$ on the right. Indeed,

$$1 - r^2 = 0.116 \longleftrightarrow V(r)/V(y) = 0.116$$

- Statistical significance: The statistical significance of a correlation depends upon its sample size $n$ as well as $r$. By tradition, we say that a correlation of $n$ points is significant if there is an $\alpha \leq 1/20 = 0.05$ chance that we would observe a correlation as strong as $r$ in any random set of $n$ points.
  This is not a particularly strong standard. Even small correlations become significant at the 0.05 level with large enough sample sizes, as shown in Figure 2.8 (right). A correlation of $r = 0.1$ becomes significant at $\alpha =$ 0.05 around $n = 300$, even though such a factor explains only $1\%$ of the variance.

Weak but significant correlations can have value in big data models involving large numbers of features. Any single feature/correlation might explain/predict only small effects, but taken together a large number of weak but independent correlations may have strong predictive power. *Maybe.* We will discuss significance again in greater detail in Section 5.3

## 4.3.1 Correlation Does Not Imply Causation!

You have heard this before: correlation does not imply causation:

- The number of police active in a precinct correlate strongly with the local crime rate, but the police do not *cause* the crime.
- The amount of medicine people take correlates with the probability they are sick, but the medicine does not cause the illness.

At best, the implication works only one way. But many observed correlations are completely spurious, with neither variable having any real impact on the other.

Still, *correlation implies causation* is a common error in thinking, even among those who understand logical reasoning. Generally speaking, few statistical tools are available to tease out whether $A$ really causes $B$. We can conduct controlled experiments, if we can manipulate one of the variables and watch the effect on

Figure 2.11: Cyclic trends in a time series (left) are revealed through correlating it against shifts of itself (right).
the other. For example, the fact that we can put people on a diet that makes them lose weight without getting shorter is convincing evidence that weight

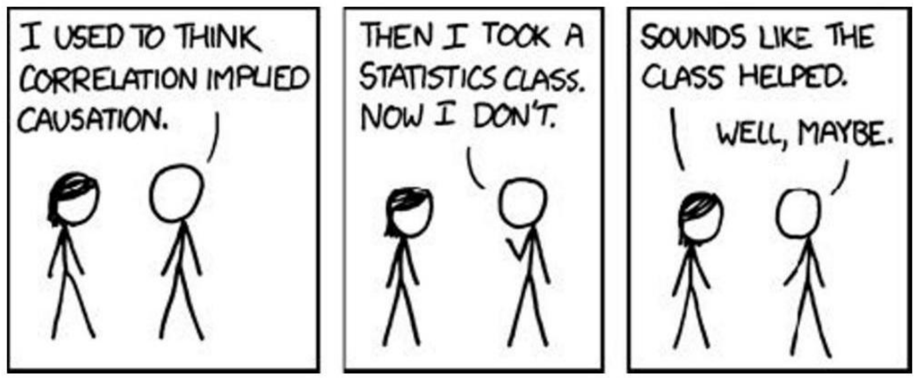



Figure 4.10: //www.xkcd
https://www.xkcd. com/552.)

does not cause height. But it is often harder to do these experiments the other way, e.g. there is no reasonable way to make people shorter other than by hacking off limbs.

### 4.3.2 Detecting Periodicities by Autocorrelation

Suppose a space alien was hired to analyze U.S. sales at a toy company. Instead of a nice smooth function showing a consistent trend, they would be astonished to see a giant bump every twelfth month, every year. This alien would have discovered the phenomenon of Christmas.

Seasonal trends reflect cycles of a fixed duration, rising and falling in a regular pattern. Many human activities proceed with a seven-day cycle associated with the work week. Large populations of a type of insect called a *cicada* emerge on a 13 -year or 17 -year cycle, in an effort to prevent predators from learning to eat them.

How can we recognize such cyclic patterns in a sequence $S$ ? Suppose we correlate the values of $S_i$ with $S_{i+p}$, for all $1 \leq i \leq n-p$. If the values are in sync for a particular period length $p$, then this correlation with itself will be unusually high relative to other possible lag values. Comparing a sequence to itself is called an *autocorrelation*, and the series of correlations for all $1 \leq k \leq n-1$ is called the *autocorrelation function*. Figure 2.11 presents a time series of daily sales, and the associated autocorrelation function for this data. The peak at a shift of seven days (and every multiple of seven days) establishes that there is a weekly periodicity in sales: more stuff gets sold on weekends.

Autocorrelation is an important concept in predicting future events, because it means we can use previous observations as features in a model. The heuristic that tomorrow's weather will be similar to today's is based on autocorrelation, with a lag of $p=1$ days. Certainly we would expect such a model to be more accurate than predictions made on weather data from six months ago (lag $p=$

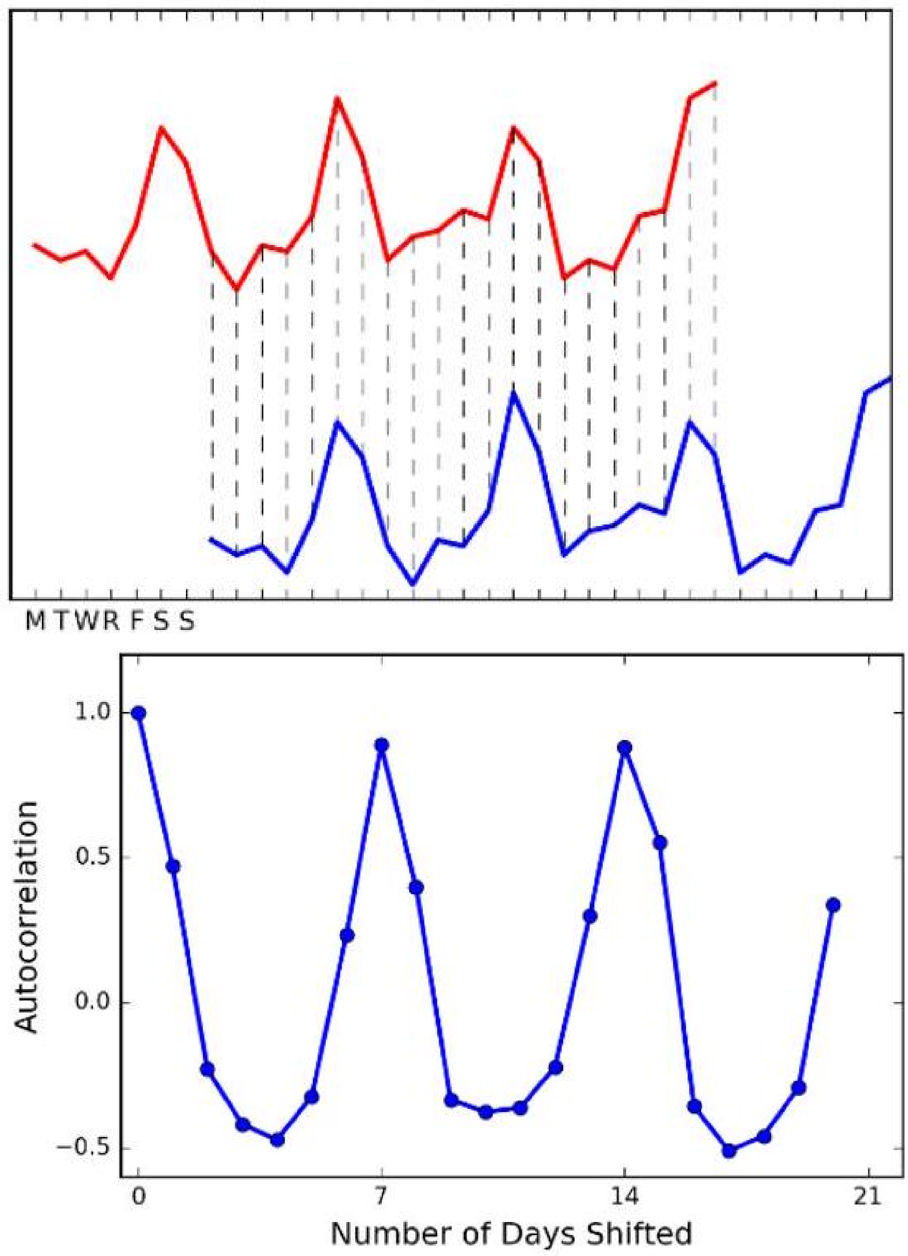



Figure 4.11: //www.xkcd
https://www.xkcd. com/552.)

180 days).

Generally speaking, the autocorrelation function for many quantities tends to be highest for *very* short lags. This is why long-term predictions are less accurate than short-term forecasts: the autocorrelations are generally much weaker. But periodic cycles do sometimes stretch much longer. Indeed, a weather forecast based on a lag of $p = 365$ days will be much better than one of $p = 180$, because of seasonal effects.

Computing the full autocorrelation function requires calculating $n-1$ different correlations on points of the time series, which can get expensive for large $n$. Fortunately, there is an efficient algorithm based on the *fast Fourier transform* (FFT), which makes it possible to construct the autocorrelation function even for very long sequences.

## 4.4 Logarithms

The *logarithm* is the inverse exponential function $y = b^x$, an equation that can be rewritten as $x = \log_b y$. This definition is the same as saying that

$$b^{\log_b y} = y.$$

Exponential functions grow at a very fast rate: consider $b = \{2^1, 2^2, 2^3, 2^4, \ldots\}$. In contrast, logarithms grow a very slow rate: these are just the exponents of the previous series $\{1, 2, 3, 4, \ldots\}$. They are associated with any process where we are repeatedly multiplying by some value of $b$, or repeatedly dividing by $b$. Just remember the definition:

$$y = \log_b x \longleftrightarrow b^y = x$$

Logarithms are very useful things, and arise often in data analysis. Here I detail three important roles logarithms play in data science. Surprisingly, only one of them is related to the seven algorithmic applications of logarithms

I present in *The Algorithm Design Manual* [
citeskiena2008algorithm]. Logarithms are indeed very useful things.

### 4.4.1 Logarithms and Multiplying Probabilities

Logarithms were first invented as an aide to computation, by reducing the problem of multiplication to that of addition. In particular, to compute the product $p = x \cdot y$, we could compute the sum of the logarithms $s = \log_b x + \log_b y$ and then take the inverse of the logarithm (i.e. raising $b$ to the $s$ th power) to get $p$, because:

$$p = x \cdot y = b^{(\log_b x + \log_b y)}.$$

This is the trick that powered the mechanical slide rules that geeks used in the days before pocket calculators.

However, this idea remains important today, particularly when multiplying long chains of probabilities. Probabilities are small numbers. Thus multiplying long chains of probability yield very small numbers that govern the chances of very rare events. There are serious numerical stability problems with floating point multiplication on real computers. Numerical errors will creep in, and will eventually overwhelm the true value of small-enough numbers.

Summing the logarithms of probabilities is much more numerically stable than multiplying them, but yields an equivalent result because:

$$\prod_{i=1}^{n} p_i = b^P, \text{ where } P = \sum_{i=1}^{n} \log_b (p_i)$$

We can raise our sum to an exponential if we need the real probability, but usually this is not necessary. When we just need to compare two probabilities to decide which one is larger we can safely stay in log world, because bigger logarithms correspond to bigger probabilities.

There is one quirk to be aware of. Recall that the $\log_2\left(\frac{1}{2}\right) = -1$. The logarithms of probabilities are all negative numbers except for $\log(1) = 0$. This is the reason why equations with logs of probabilities often feature negative signs in strange places. Be on the lookout for them.

Logarithms and *Ratios* Ratios are quantities of the form $a/b$. They occur often in data sets either as elementary features or values derived from feature pairs. Ratios naturally occur in normalizing data for conditions (i.e. weight after some treatment over the initial weight) or time (i.e. today's price over yesterday's price).

But ratios behave differently when reflecting increases than decreases. The ratio $200/100$ is $200\%$ above baseline, but $100/200$ is only $50\%$ below despite being a similar magnitude change. Thus doing things like averaging ratios is committing a statistical sin. Do you *really* want a doubling followed by a halving to average out as an increase, as opposed to a neutral change?

One solution here would have been to use the geometric mean. But better is taking the logarithm of these ratios, so that they yield equal displacement, since $\log_2 2 = 1$ and $\log_2(1/2) = -1$. We get the extra bonus that a unit ratio maps to zero, so positive and negative numbers correspond to improper and proper ratios, respectively.

A rookie mistake my students often make involves plotting the value of ratios instead of their logarithms. Figure 2.12 (left) is a graph from a student paper, showing the ratio of new score over old score on data over 24 hours (each red dot is the measurement for one hour) on four different data sets (each given a row). The solid black line shows the ratio of one, where both scores give the same result. Now try to read this graph: it isn't easy because the points on the left side of the line are cramped together in a narrow strip. What jumps out at you are the outliers. Certainly the new algorithm does terrible on 7UM917 in the top row: that point all the way to the right is a real outlier.

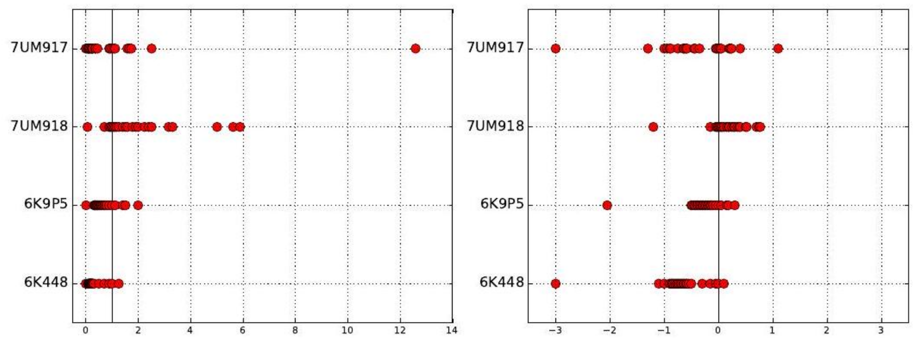



Figure 4.12: Plotting ratios on a scale cramps the space allocated to small ratios relative to large ratios (left). Plotting the logarithms of ratios better represents the underlying data (right).

Except that it isn't. Now look at Figure 2.12 (right), where we plot the logarithms of the ratios. The space devoted to left and right of the black line can now be equal. And it shows that this point wasn't really such an outlier at all. The magnitude of improvement of the leftmost points is much greater than that of the rightmost points. This plot reveals that new algorithm generally makes things better, only because we are showing logs of ratios instead of the ratios themselves.

### 4.4.2 Logarithms and Normalizing Skewed Distributions

Variables which follow symmetric, bell-shaped distributions tend to be nice as features in models. They show substantial variation, so they can be used to discriminate between things, but not over such a wide range that outliers are overwhelming.

But not every distribution is symmetric. Consider the one in Figure 2.13

(left). The tail on the right goes much further than the tail on the left. And we are destined to see far more lopsided distributions when we discuss power laws, in Section 5.1.5 Wealth is representative of such a distribution, where the poorest human has zero or perhaps negative wealth, the average person (optimistically) is in the thousands of dollars, and Bill Gates is pushing $100 billion as of this writing.

We need a normalization to convert such distributions into something easier to deal with. To ring the bell of a power law distribution we need something non-linear, that reduces large values to a disproportionate degree compared to more modest values.

The logarithm is the transformation of choice for power law variables. Hit your long-tailed distribution with a log and often good things happen. The distribution in Figure 2.13 happened to be the *log normal* distribution, so taking

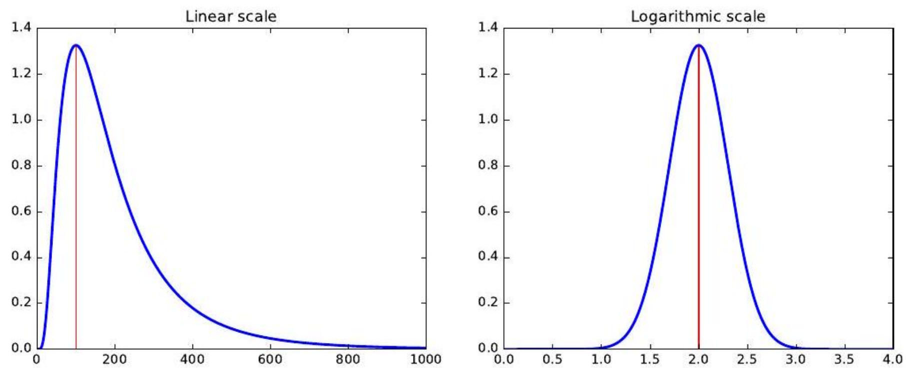



Figure 4.13: Hitting a skewed data distribution (left) with a $\log$ often yields a more bell-shaped distribution (right).

the logarithm yielded a perfect bell-curve on right. Taking the logarithm of variables with a power law distribution brings them more in line with traditional distributions. For example, as an upper-middle class professional, my wealth is roughly the same number of logs from my starving students as I am from Bill Gates!

Sometimes taking the logarithm proves too drastic a hit, and a less dramatic non-linear transformation like the square root works better to normalize a distribution. The acid test is to plot a frequency distribution of the transformed values and see if it looks bell-shaped: grossly-symmetric, with a bulge in the middle. That is when you know you have the right function.

War Story: Fitting Designer *Genes* The word *bioinformatician* is life science speak for "data scientist," the practitioner of an emerging discipline which studies massive collections of DNA sequence data looking for patterns. Sequence data is very interesting to work
with, and I have played bioinformatician in research projects since the very beginnings of the human genome project.

DNA sequences are strings on the four letter alphabet $\{A, C, G, T\}$. Proteins form the stuff that we are physically constructed from, and are composed of strings of 20 different types of molecular units, called amino acids. Genes are the DNA sequences which describe exactly how to make specific proteins, with the units each described by a triplet of $\{A, C, G, T\}$ s called *codons*.

For our purposes, it suffices to know that there are a huge number of possible DNA sequences describing genes which *could* code for any particular desired protein sequence. But only one of them is used. My biologist collaborators and I wanted to know why.

Originally, it was assumed that all of these different synonymous encodings were essentially identical, but statistics performed on sequence data made it clear that certain codons are used more often than others. The biological conclusion is that "codons matter," and there are good biological reasons why this should be.

We became interested in whether "neighboring pairs of codon matter." Perhaps certain pairs of triples are like oil and water, and hate to mix. Certain letter pairs in English have order preferences: you see the bigram $gh$ far more often than $hg$. Maybe this is true of DNA as well? If so, there would be pairs of triples which should be underrepresented in DNA sequence data.

To test this, we needed a score comparing the number of times we actually see a particular triple (say $x = CAT$ ) next to another particular triple (say $y = GAG$) to what we would expect by chance. Let $F(xy)$ be the frequency of $xy$, number of times we actually see codon $x$ followed by codon $y$ in the DNA sequence database. These codons code for specific amino acids, say $a$ and $b$ respectively. For amino acid $a$, the probability that it will be coded by $x$ is $P(x) = F(x)/F(a)$, and similarly $P(y) = F(y)/F(b)$. Then the expected number of times of seeing $xy$ is

$$\textit{Expected}\,(xy) = \left(\frac{F(x)}{F(a)}\right)\left(\frac{F(y)}{F(b)}\right)F(ab)$$

Based on this, we can compute a codon pair score for any given hexamer $xy$ as follows:

$$CPS(xy) = \ln\left(\frac{\textit{Observed}\,(xy)}{\textit{Expected}}(xy)\right) = \ln\left(\frac{F(xy)}{\frac{F(x)F(y)}{F(a)F(b)}F(ab)}\right)$$

Taking the logarithm of this ratio produced very nice properties. Most importantly, the sign of the score distinguished over-represented pairs from underrepresented pairs. Because the magnitudes were symmetric ( +1 was just as impressive as -1 ) we could add or average these scores in a sensible way to give a score for each gene. We used these scores to design genes that should be bad for viruses, which gave an exciting new technology for making vaccines. See the chapter notes (Section 2.6) for more details.

| Fr. Dep. | Score | | Fr. Ind. | Score |
|---|---|---|---|---|
| CATAGG | -1.74 | | GGGGGG | -1.01 |
| TCTAGC | -1.61 | | CCCCCC | -0.95 |
| GTTAGG | -1.58 | | GGCGCC | -0.66 |
| GCTAGT | -1.48 | | GGGGGT | -0.63 |
| CCTAGT | -1.44 | | CGGGGG | -0.59 |
| GGTAGG | -1.41 | | AGGGGG | -0.58 |
| CTTAGG | -1.40 | | CACGTG | -0.58 |
| ACTAGC | -1.38 | | ACCCCC | -0.56 |
| GCTAGC | -1.37 | | GGGCCC | -0.56 |
| GCTAGA | -1.36 | | CCCCCT | -0.53 |
| CCTAGC | -1.35 | | CGCCCC | -0.52 |
| GATAGG | -1.35 | | CCCCCG | -0.51 |